

Misleading Names Can Distort Written Reasoning

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Abstract

Kudo et al. (2026) showed that language models can compute answers while generating intermediate steps in arithmetic. Building on this work, we ask whether misleading words can pull written reasoning away from the correct computation. We change variable names in arithmetic problems and Python programs while leaving the computations unchanged. Writing out calculations reduces answers suggested by misleading names, but errors remain possible in both settings. In code, the correct value is recoverable from internal activity with 95.5–99.7% accuracy immediately before the model writes a conflicting value suggested by the name. Changing worked examples to show each source operation alongside its result sharply reduces these mistakes. Adding text of the same length leaves large errors, and extra arithmetic helps only partly. Correct information can therefore be available inside a model without appearing in its written reasoning. Demonstrating the calculation behind each step improves resistance to misleading cues in the tested tasks.

1 Introduction

Kudo et al. (2026) provide evidence that language models compute intermediate answers as they generate arithmetic reasoning. Their controlled tasks make it possible to track which values become readable inside a model and how changing its internal activity affects the final answer. This connects written calculations to the computation that develops during generation. We build on their task and methods to study a conflict that arithmetic alone does not introduce. What happens when a familiar word suggests an answer different from the one the equations require?

Consider a variable whose value is 6 but whose name is four. Renaming it does not change the mathematical problem, yet a model can answer 4. The same conflict occurs in code when a variable

named `sum_all` stores a list’s length. For `[5, 3]`, the code computes 2 while the name suggests 8. These examples separate two identifiable values rather than treating any incorrect response as evidence of an effect of wording. They let us ask whether written steps follow the computation or the suggestion carried by the name (Figure 1).

We introduce this conflict in both Kudo et al.’s arithmetic task and short Python programs. Matched problems differ only in a variable name. We first compare final answers with and without written calculations, then inspect what happens when a written value follows the misleading name. A separately trained predictor reads the correct digit from internal activity immediately before the model’s actual wrong write. Finally, we change the worked examples to test whether showing the relevant operation beside its result reduces these errors.

Writing arithmetic calculations reduces name-suggested answers, but neither a correct written value nor a readable internal value guarantees a correct output. In code, the correct digit is recoverable on actual wrong writes with 95.5–99.7% accuracy. Demonstrations that show the source expression substantially reduce those writes, including on fresh programs with independently screened names. Length-matched text leaves large effects. Together, the matched renaming tests and error-conditioned readouts show where written reasoning can depart from correct information. The demonstration comparison identifies a way to reduce that departure.

2 Experimental design

2.1 A shared conflict with two different tasks

Both tasks pair an ordinary-name problem with a misleading-name version of the same computation. The inputs, operators, dependencies and requested value remain fixed. We call an output that matches

(a) Arithmetic

```
pen = 1 + cup  
cup = 2 + 3  
pen = ?
```

pen → four

Computed value 6

Name suggests 4

(b) Python code

```
xs = [5, 3]  
v = len(xs)
```

v → sum_all

Computed value 2

Name suggests 8

Figure 1: Renaming preserves the computation but suggests a different answer.

the name’s incorrect suggestion a *name error*. *Excess name errors* measures how much renaming increases that output. It subtracts the ordinary-name twin’s rate for the same suggested value from the misleading-name rate. The subtraction matters because a model can guess that digit without any misleading word. For example, a model that answers 4 equally often in both versions has zero excess, even if its overall accuracy is low.

Arithmetic. We reimplement Kudo et al.’s task of single-digit addition and subtraction. The main level requires two operations and presents the queried variable’s equation before the equation needed to evaluate it. In `pen=1+cup, cup=2+3; pen=?`, the queried value is 6. Replacing `pen` with `four` changes only the name. The misleading digit is absent from every input and variable value, so answering 4 cannot be attributed to copying an existing digit. We also test names that denote the correct digit and rename intermediate variables separately. All names occupy one token, preserving token positions within matched pairs.

The arithmetic panel contains thirteen checkpoints, including pretrained, instruction-tuned and derived models. Two pretrained Llama models cover all five task levels. Each level has 2,000 matched problems; demonstrations use ordinary names disjoint from test names. The main behavioral table applies the planned 90% ordinary-name accuracy floor under written calculations. Results below that floor remain visible in the supporting tables. Appendix A gives the levels, model list and tokenizer checks.

Code. Each task is a short Python function over distinct input digits. Its assignments compute length, sum, maximum or minimum, then apply integer arithmetic. All variable and return values are single digits. The main level has three dependent assignments. For a length assignment, a sum-like

name suggests the sum of the list while the executable value remains its length. The suggested value differs from every input element and from the correct values. We evaluate seven instruction-tuned models at levels 3 and 5, with 1,000 matched problems per target, and add CodeLlama-34B as a scale check. The arithmetic and code panels overlap in model families but differ in checkpoints and prompts; they are separate experiments rather than a pooled model sample. The length-to-sum comparison contains 285 matched programs per demonstration set within the larger task pool.

The original code design planned to retain identifiers that evoke their intended operation in at least 80% of models. The runs instead retained the full identifier table. We disclose that selection deviation and report the results for those names. A subsequent replication screens fresh names before observing traces and uses 200 new computations. It tests whether the behavioral format effect generalizes to independently selected names; it does not retroactively change the original cohort (Appendix T).

2.2 What the model is asked to write

In arithmetic, *answer-only* prompting requests the final digit. The *calculation* condition asks the model to restate and evaluate the equations before answering, for example `cup=2+3=5, four=1+5=6; Answer: 6`. This is the written chain of thought. An answer-only response can still involve internal computation; the comparison concerns the output the prompt requests.

In code, a *values-only trace* writes each assignment’s value, such as `sum_all=2`, before the return value. An *expression-and-value trace* shows the source operation as well, such as `sum_all=len(xs)=2`. Worked examples demonstrate these formats while the test program and Trace: prefix remain fixed. Other conditions use

an interactive Python transcript, code comments with values, or free prose. These formats make different information explicit; we do not treat all written steps as the same intervention.

Main behavior uses three fixed sets of three worked examples, with demonstration seeds 7, 11 and 13. Decoding is greedy over the full vocabulary. Parse failures remain in denominators. We report ordinary-name accuracy beside name errors because avoiding a particular wrong value does not establish successful computation.

2.3 Outputs, readouts and interventions

Behavioral measurements score what the model generates. A *linear probe* is a separately trained predictor of the correct digit from one internal hidden state. It measures which information can be recovered at a position and layer. The language model does not use this predictor to generate its answer. *Activation patching* replaces one problem’s internal activity with activity from its ordinary-name twin and measures the resulting output change. These three measurements answer different questions.

Probes train on separate ordinary-name computations with three initialization seeds. Controls assign a random digit to each name to measure name identity (Hewitt and Liang, 2019). The central code analysis uses five-fold layer selection that excludes the held-out computation. It reads the = immediately before the model’s actual wrong value; the generated and cached token prefixes must agree through that position. No answer digit is included. Arithmetic measurements mainly supply a correct calculation token by token. We separately test freely generated calculations and report whether the probe generalizes to their ordinary-name states.

For reliable behavioral effects, the sign must agree across all three demonstration sets and a 95% cluster-bootstrap interval must exclude zero. Small-effect claims require a 90% interval entirely inside ± 2 percentage points. Bootstrap clusters keep matched names and demonstration repeats together. Controls involving repeated computations cluster on the underlying arithmetic or program rather than the spelling of its variables. Intervals are pointwise. They describe the tested cells, not a multiplicity-corrected result over independent model replications.

3 Misleading names can affect written reasoning

3.1 Writing calculations improves resistance in arithmetic

Without calculations, a misleading name increases answers matching the digit it suggests in several arithmetic conditions (Figure 2a). For pretrained Llama-3.1-8B at the main level, a correct number-word name improves accuracy by +4.8 points, while a misleading name adds +1.2 points of name errors on the queried variable. The matched baseline discounts digit guesses when answer-only accuracy is low.

Writing calculations reduces excess name errors reliably in 10 of 44 paired comparisons. With calculations, 43 of those 44 added-error intervals fit within two points of zero. These are different statements. Before calculations, 31 comparisons already meet that bound; the remaining small effects cannot all be counted as demonstrated improvements. A sensitivity analysis clustering repeated computations retains the same ten reliable reductions (Appendix G).

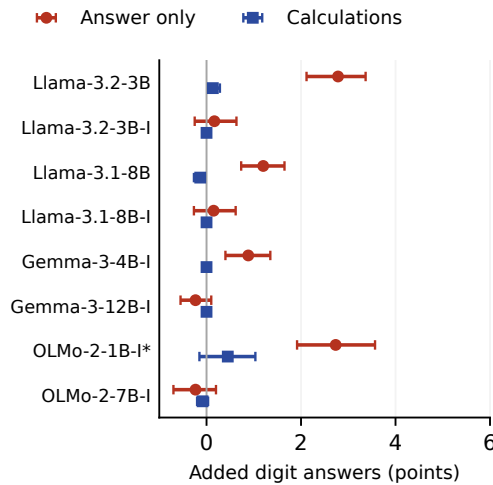
The exception is OLMo-2-1B-Instruct’s intermediate-variable condition, with +3.2 added-error points [+2.5, +3.9]. Its ordinary-name calculation accuracy is 47%, compared with 77% under letter names. Accuracy also remains sensitive to names in other models even when excess name errors are small. Across the calculation conditions, 15 accuracy contrasts remain reliable. Resistance to the misleading digit and general arithmetic accuracy must therefore be reported separately.

3.2 A correct written value does not guarantee a correct answer

Among matched arithmetic generations that both state the intermediate variable’s true value, OLMo-2-1B still shows +3.6 added-error points [+2.6, +4.7] over 2739 paired observations. This is a conditional comparison of observed calculations. It shows that stating a correct value is compatible with a final name error, without estimating the causal effect of stating that value.

A stricter control holds an entire completely correct generated calculation fixed and changes only the final query and answer format. Neither tested Llama model produces a name-suggested digit in those continuations. OLMo-2-1B does. An unnamed final query with digit-only output

(a) Arithmetic, queried variable



I: instruction-tuned. *Calculation accuracy below 90%.

(b) Code, length named as sum

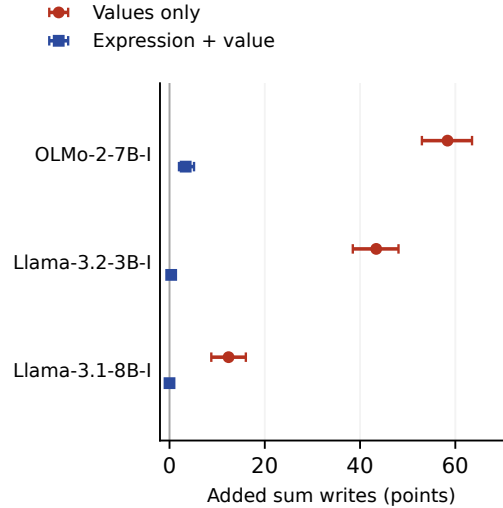


Figure 2: Selected arithmetic and code name-error contrasts (95% intervals); scales differ.

produces 12.8 added-error points [9.1, 16.6], despite the preceding correct calculation. Repeating the queried name reduces the estimate to 0.5 points, but also raises ordinary-name accuracy from 78.0% to 96.2%. The final format changes competence as well as susceptibility. These results establish that a correct calculation is not sufficient for reliable final answering (Appendix H).

Name errors also need not copy the last digit in the calculation. Across 4110 generated calculation name errors, only 13% end the calculation segment with the name-suggested digit. A trailing instance of that digit is therefore not necessary for the failure.

3.3 Code traces can write the name’s suggested operation

The same matched conflict produces larger effects in a particular code format. When worked examples list values alone, length-computing programs acquire +58.4, +43.4 and +12.4 points of excess sum writes in OLMo-2-7B, Llama-3.2-3B and Llama-3.1-8B, respectively. Across ordinary-name programs at this level, their final-answer accuracies are 95.7, 97.4 and 99.6%. All three are instruction-tuned checkpoints. The suggested sum is not their usual error on a length step. On parsed incorrect ordinary-name writes, sums account for only 14–32% of errors. Maximum values and off-by-one lengths are more common.

The effect depends on the task and output format. Direct-answer effects at the same code level

range from -0.2 to +3.1 points, much smaller than the largest values-only effects. CodeLlama-7B’s stronger conflict occurs on a maximum-computed variable, while CodeGemma-7B shows that pairing at level 3 (Appendix K). Thus written steps can reduce a conflict in arithmetic and expose a conflict in a code trace. The shared result is sensitivity to misleading names, rather than a universal direction of change caused by showing work.

4 Correct values can precede incorrect writes

4.1 Read the state immediately before the observed failure

Does a wrong written value mean the correct value is unavailable inside the model? The code error cases provide a direct test. On actual sum writes where the code requires length, independently selected probes recover the correct digit with 95.5–99.7% accuracy immediately before the wrong value is generated (Figure 3). The observed error cohorts contain 503 writes from 193 computations for OLMo-2-7B, 372 from 169 for Llama-3.2-3B, and 106 from 45 for Llama-3.1-8B. Repeated demonstrations remain clustered; the write counts are not counts of independent programs.

The readout becomes more accurate during the trace. Before any trace token is generated, correct-digit accuracy on those same future error cases is 38.4%, 26.4% and 56.0%, respectively. Immediately before the wrong value, it is 96.4%, 95.5%

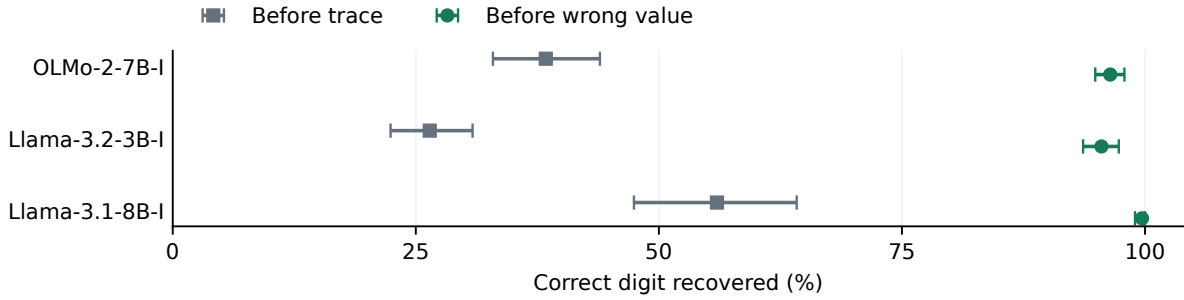


Figure 3: Correct-digit readouts on the same wrong writes (95% intervals). I: instruction-tuned.

and 99.7%. Paired improvements are 58.1 [52.6, 63.4], 69.1 [64.5, 73.4] and 43.7 [35.4, 52.4] points. Both positions use independent layer selection on the same computations. The comparison places the discrepancy at the point of writing, rather than merely finding a high score somewhere in the network (Appendix U).

The training and measurement controls are central to interpreting this result. The predictors train on 2,000 separate ordinary-name programs. Renamed versions of one computation cannot cross training or evaluation splits. Five-fold selection chooses a layer using ordinary-name labels from other folds; the held-out computation and its misleading writes do not determine the choice. The cached prefix matches the actual generation through the = in all 2,565 audited generations. The next digit has not yet been written, so a predictor cannot obtain it from that future token. Intervals condition on the fixed layer selectors and keep demonstrations of one computation together.

The readout is a measurement, not a second answer produced by the language model. It shows that the correct value can be recovered while the model is about to write another value. It does not show which feature the model uses to select that output, or establish that the readout alone could correct it. This distinction is precisely why correct internal information cannot certify a written explanation.

4.2 Arithmetic readouts support a narrower comparison

The arithmetic analysis reproduces Kudo et al.’s timing of readable intermediate values under supplied correct calculations. Llama-3.2-3B’s correct-digit probe accuracy reaches 97.9% just before the value is written, while name-suggested predictions fall to zero. This demonstrates the readout method

on the arithmetic task, but the supplied calculation is a different context from an actual generated wrong code write.

When arithmetic probes transfer to freely generated calculations, their reliability depends on the model. Retraining on generated ordinary-name calculations gives 96.3–100.0% held-out accuracy for Llama-3.2-3B, but only 50.3–80.1% for OLMo-2-1B. OLMo fails the 90% calibration criterion in every tested cell, and Llama’s name-error cases are rare. These measurements cannot establish the same strong error-conditioned result as the code analysis (Appendix F).

Across nine arithmetic models, the association between misleading-name probe accuracy and name errors reverses when OLMo is removed. Arithmetic competence is a competing explanation for that descriptive association. We therefore base the central internal claim on the code readouts before observed failures, with explicit training and selection controls, rather than inferring a mechanism from cross-model correlations.

4.3 Replacing activity changes which value is written

Patching supplies a separate intervention on the output. In the two code models with aligned name-site twins, replacing early name-token activity with the ordinary-name program’s activity removes 97% and 100% of name errors. Replacement immediately before the value is weak in early layers and strong from the middle of the network onward. Replacing every layer at that position removes 97–100% of name errors. The all-layer neutral-word control changes at most 1.0% of initially correct outputs (Appendix M).

The arithmetic interventions also identify positions sensitive to misleading names, but can disrupt correct answers. Every patch replaces a whole

hidden-state vector. Removing a name-suggested answer can produce a different wrong answer; it is not automatically a correction. The interventions show that activity at these positions affects generation, without isolating a name feature or one common mechanism across the two tasks.

5 Showing the operations reduces errors

5.1 Change the demonstrations, then inspect the same programs

The strongest practical comparison keeps the program, input, model and demonstration identities fixed. Replacing $v=2$ examples with $v=\text{len}(xs)=2$ connects the reported value to the source operation. Added sum writes fall to +3.4, +0.4 and +0.0 points in OLMo-2-7B, Llama-3.2-3B and Llama-3.1-8B, respectively. Figure 4 compares the source expressions with additional text and numeric elaboration on the same programs.

Could the additional text alone explain the improvement? Descriptive annotations matched to the expression examples’ token counts leave 56.6 and 49.6 points of excess in OLMo-2-7B and Llama-3.2-3B. Those effects remain much larger than under source-expression examples. The matching is in the demonstrated outputs; the test program and trace prefix are unchanged.

Valid numeric elaboration, such as $v=2+0=2$, tests whether any added arithmetic produces the benefit. It helps partly, leaving 41.3 and 14.9 points of excess in the same two models. Both exceed the corresponding source-expression effects. The paired changes and intervals are in Appendix R. These controls favor a role for the content of the operation shown in the examples, while changing several aspects of the written context. They do not identify a single responsible representation.

Interactive Python transcripts and values written as code comments also reduce errors (Appendix Table 25). Source expressions are one effective demonstration format for these tasks.

5.2 The format contrast replicates with fresh names

The original code cohort includes names retained contrary to the planned selection rule. We therefore screen 16 fresh sum-family candidates by their highest-scoring operation continuation before testing traces. Of these, 14 pass the 80% panel criterion. We freeze the first 3 qualifying names

and cross them with 200 new length-computing programs. The computations are absent from the original training and test data and every demonstration set.

With values-only examples, added sum writes are 91.1, 61.5 and 16.7 points in OLMo-2-7B and the two Llama models. Expression-and-value examples reduce these to 0.8 points in OLMo and zero observed excess in both Llamas. The pre-specified CodeGemma-7B control has zero observed excess in both formats. Figure 5 shows that the contrast holds across all three selected names. Intervals cluster repeated demonstrations of each computation; pooled intervals also cluster names (Appendix T). The replication supports the format contrast on independently selected names; its changed names and programs make magnitude differences from the original cohort descriptive. Probes and patches still concern the original identifiers.

5.3 Arithmetic controls separate equations from repeated names

Arithmetic calculations both evaluate equations and repeat variable names. We cross those properties in worked examples for pretrained Llama-3.2-3B, pretrained Llama-3.1-8B and OLMo-2-1B-Instruct. Outputs show named equations, unnamed numeric equations, named values alone, or unnamed values alone. Neutral notes match full prompt token counts. Generated response lengths are measured rather than forced to match.

No model meets the common accuracy criterion across all four formats; all use 16 examples after separate calibration. In the two Llama models, however, named and unnamed equations both reach 99–100% ordinary-name accuracy. Those four role comparisons pass the held-out competence check, and excess name errors remain near zero without repeated names. Name repetition is therefore unnecessary for high accuracy and low name errors in these tested equation conditions. Values-only comparisons fail the competence criterion, so they cannot isolate an equation effect at matched accuracy. OLMo’s unnamed equations also reduce accuracy (Appendix H).

Together, the controls show why the content and format of written steps deserve separate evaluation. Explicit source operations improve the code trace contrast; arithmetic equations can remain effective without repeated names. Neither result establishes a format-independent causal account of reasoning.

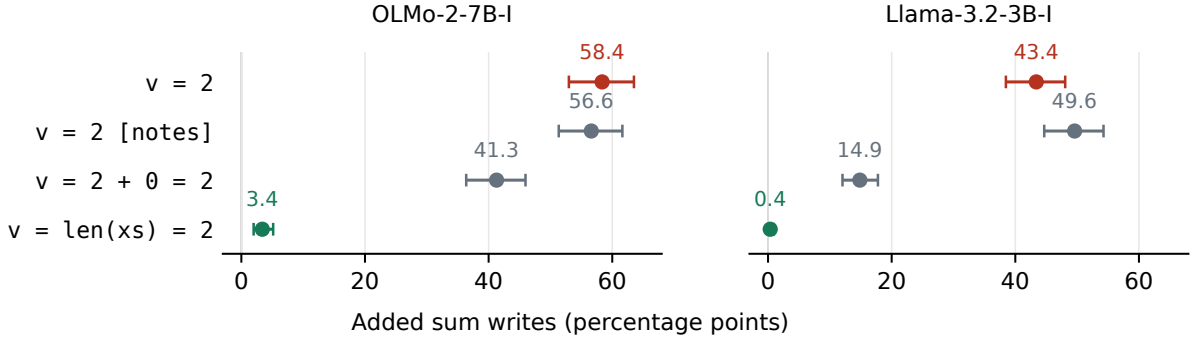


Figure 4: Length-computing code: format controls (95% intervals). [notes]: neutral text.

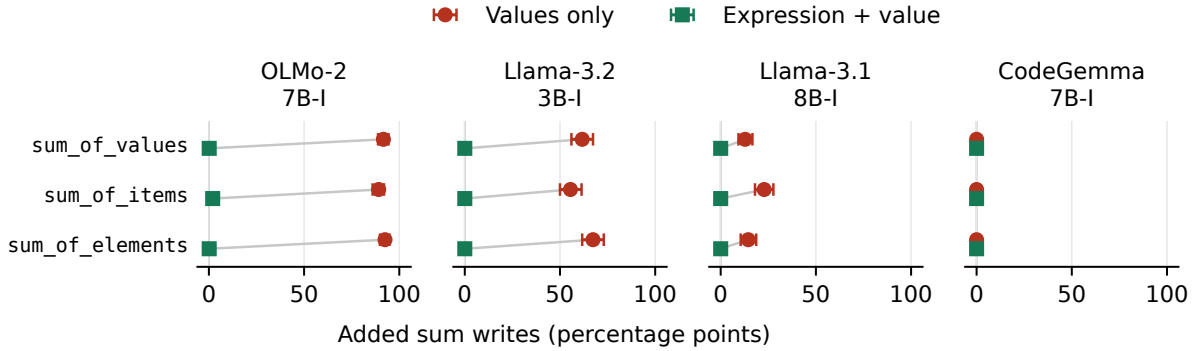


Figure 5: Fresh names on 200 new length computations (95% intervals). I: instruction-tuned.

6 Related work

Computation during reasoning. Kudo et al. (2026) track arithmetic values with probes and activation interventions, finding that answers develop during the calculation. We retain their controlled arithmetic and probe recipe, then introduce a name with a competing answer and extend the conflict to programs. Our code timing comparison concerns the same observed wrong writes before and during a trace. Work on positional readouts and shortcut solutions also motivates testing what a readout establishes (Liu, 2026; Prabhakar et al., 2024).

Names and misleading wording. Identifier changes affect code comprehension and robustness (Gao et al., 2023; Le et al., 2025; Lam et al., 2025; Le et al., 2026; Orvalho and Kwiatkowska, 2025). Our tasks provide an executable correct value and a distinct value suggested by the name. The matched comparison measures which incorrect value changes, rather than attributing every accuracy loss to the same influence. Stroop and cue-conflict studies supply a related experimental logic (Stroop, 1935; Geirhos et al., 2019; Wang, 2026; Hu et al., 2026); the analogy does not estab-

lish a shared human and model mechanism.

Written reasoning and internal information. Faithfulness studies show that explanations can omit influences on outputs or react differently to perturbations (Turpin et al., 2023; Lanham et al., 2023; Chen et al., 2025). Our central discrepancy is narrower and directly measured. A correct digit is recoverable immediately before the model writes a conflicting digit in a code trace. Probe controls distinguish decodability from name identity, while patching tests an intervention separately (Hewitt and Liang, 2019; Zhang and Nanda, 2024). Changing the demonstrated trace then connects this discrepancy to an output format that can reduce the error.

7 Discussion and conclusion

The experiments connect three observations that evaluations often treat as interchangeable. A model can state a correct value in a calculation, have the correct value readable from internal activity, and produce a final answer. Evidence for one does not establish the others. Correct arithmetic calculations can precede a name-suggested final answer. Correct code values can be recovered

immediately before actual wrong writes. Asking for written steps therefore gives useful evidence about a computation, but the steps themselves still require checking.

The results suggest a practical evaluation procedure. Preserve the computation, rename a variable to suggest a distinct wrong value, and score both the written intermediate values and the final answer. Report overall accuracy alongside the increase in the suggested error. Repeat the comparison across demonstration sets and output formats. This distinguishes resistance to the misleading cue from general task competence and avoids treating a small excess as proof of success.

Demonstrating the source operation alongside its result is an effective change to the code examples. It leaves the program and model fixed and sharply reduces name-suggested writes, including in the independently screened replication. Additional text of the same length does not reproduce the benefit, and numeric elaboration reproduces only part of it. The arithmetic controls show that repeated names are unnecessary in the tested competent Llama equation formats. These findings support evaluations of what is written at each step, rather than a single category of “reasoning enabled.”

Scale and natural-code checks bound the result. CodeLlama-34B’s maximum-to-sum excess is +1.8 points [+0.7, +3.1]; its interval crosses the two-point small-effect margin. In the 89-function CRUXEval check (Gu et al., 2024), accuracy contrasts cover zero on the 84 functions that use the renamed variable downstream. The tested formats are direct answers and prose (Appendix N).

The next question is whether this demonstration change transfers to larger programs and natural explanations. A useful test would compare formats on independent functions with enough correct ordinary-name performance, then audit the states before actual wrong writes. A more selective intervention would be needed to test whether the information recovered by the probe influences the generated digit. The present whole-state patches do not establish that causal link. The results show that a correct internal readout can precede a wrong written value, and that examples showing the operation reduce this failure in the tested tasks.

Limitations

- **Tasks and formats.** Inputs are synthetic arithmetic and short programs with single-digit values. The largest code effects use a values-only trace; interactive Python and comment formats show smaller effects. A 89-function CRUXEval check tests direct-answer and prose accuracy, not the synthetic trace. Its intervals cover zero and do not establish equivalence within the two-point margin. Models reach 34B, but the arithmetic and code panels differ in checkpoints and do not cover all model families.
- **Readouts and interventions.** The central code probes cover three models. They establish recoverable values, not causal use. Arithmetic generated-calculation probes are unreliable for OLMo and have rare Llama errors. Patches replace mixed representations and may produce another wrong answer. Neither the probes nor the patches identify one mechanism shared across the tasks.
- **Selection and competence.** The original code identifiers depart from the planned meaning gate; the independently screened replication tests new names and computations. Accuracy changes accompany several arithmetic format controls. Conditional correct-calculation cohorts do not estimate the effect of making a calculation correct. Intervals condition on fitted selectors and the tested demonstrations; related checkpoints and repeated roles are not independent models.

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	J. Ridley Stroop. 1935. Studies of interference in serial verbal reactions. <i>Journal of Experimental Psychology</i> , 18(6):643–662.	744
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	Team OLMo. 2024. 2 OLMo 2 furious. <i>arXiv preprint arXiv:2501.00656</i> .	746
		747
	Miles Turpin, Julian Michael, Ethan Perez, and Samuel Bowman. 2023. Language models don’t always say	

Name	Problem (answer: 6)
Neutral	pen=1 + cup, cup=2 + 3; pen=?
Congruent	six=1 + cup, cup=2 + 3; six=?
Incongruent	four=1 + cup, cup=2 + 3; four=?

Table 1: Matched names preserve the arithmetic. The misleading digit, 4, occurs nowhere else.

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A Arithmetic task and reproducibility

A problem contains single-digit variables, addition or subtraction, and a query. The five levels vary dependency order, depth and the presence of an unused equation (Table 2). Three same-level worked examples precede each test problem. The evaluated models are Gemma-3-12B-it, Gemma-3-4B, Gemma-3-4B-it, Llama-3.1-8B, Llama-3.1-8B-Instruct, Llama-3.2-3B, Llama-3.2-3B-Instruct, OLMo-2-1B-Instruct, OLMo-2-7B-Instruct, plus four derived checkpoints (Appendix B.3). Demonstrations use ordinary names for every word-name condition, with a reserved word pool disjoint from test names. Generation is greedy; only the test variable name changes between matched conditions.

We reimplement Kudo et al. (2026)’s task and probe recipe rather than using their code. The shared choices are single-digit outputs, same-level

Level	Problem (neutral names)	Steps	Held	Distr.
1	pen=1 + cup, cup=2; pen=?	1	1	0
2	pen=2 + 3, cup=1 + pen; cup=?	2	0	0
3	pen=1 + cup, cup=2 + 3; pen=?	2	1	0
4	pen=1 + cup, cup=2 + 3, box=4 + 5; pen=?	2	1	1
5	pen=1 + cup, cup=2 + box, box=1 + 2; pen=?	3	2	0

Table 2: The five levels of Kudo et al. (2026): additions or subtractions needed (*Steps*), variables unresolved when first read (*Held*), and unused equations (*Distr.*). At level 1 the intermediate value is stated rather than computed.

demonstrations, a 10-way linear residual-stream classifier, full-batch SGD at 10^{-3} for 10,000 epochs, and their span-by-layer-window patching design. For replication we train on letter names. The Llama-3.2-3B value-writing positions agree with their Table 3: equation 5 for the queried variable and 2 for the intermediate. Our additions are the conflicting names, matched base-lines, demonstration-set claim rule and equivalence bound.

Pre-chain probe maxima are 30.4 and 47.9, compared with 17.8 and 33.2 in Kudo et al. (2026). The value-writing positions agree, while the accuracy difference may reflect training details.

A.1 Tokenizer checks

We test candidate names in five tokenizer contexts and retain words that form one token without merging with a neighbor. For Llama 3, all ten number words and 95 of 112 neutral candidates pass. Gemma and OLMo use the same lists; run-time checks confirm equal token lengths and positions within matched groups. Spaces around binary operators prevent merges such as 1-two. No instance in the reported runs is excluded by the layout check.

Data, compute and software. Each level has 2,000 matched sets, each rendered in every condition, and 10,000 separately generated problems for training probes; every SGD probe is trained with 3 seeds. Every run here is inference and the probes are linear; the three adapter checkpoints of Appendix B.3 were trained in a separate project, and that compute is not counted below. Combined allocation time across the arithmetic and code experiments totals 222.5 GPU-hours on NVIDIA A30, H100 and H200 cards, an upper bound including idle and shared workers. The models are from

the Llama 3 (Grattafiori et al., 2024), Gemma 3 (Gemma Team, 2025) and OLMo 2 (Team OLMo, 2024; Walsh et al., 2025) families, with the reasoning-tuned checkpoint from Bercovich et al. (2025), each used under its release licence for research. Models run in PyTorch (Paszke et al., 2019) with Transformers (Wolf et al., 2020), and the probes are trained in PyTorch; the L-BFGS re-fit of Appendix C.1 uses scikit-learn (Pedregosa et al., 2011), and the mixed-effects check uses statsmodels (Seabold and Perktold, 2010). Llama checkpoints use the Llama 3.1 or 3.2 Community License, Gemma uses the Gemma Terms of Use, OLMo uses Apache 2.0, and Nemotron uses the NVIDIA Open Model License. We use released weights for research inference and do not redistribute them. Versions are PyTorch 2.11.0 (CUDA 12.9), Transformers 5.14.1, NumPy 2.3.5, scikit-learn 1.9.1 and SciPy 1.18.0.

B Full arithmetic results and controls

B.1 Equivalence and accuracy floor

Following preregistration Amendment 3, equivalence requires a 90% cluster-bootstrap interval wholly inside the ± 2 -point margin (two one-sided 5% tests). We resample matched sets with their demonstration-seed replicates together, using 4,000 bootstrap draws. Other behavioral intervals remain at 95%. For CoT accuracy, 61 of 88 facilitation and interference contrasts meet the same equivalence bound. Of the 15 reliable accuracy contrasts, 5 are also equivalent within the margin and 10 are not; a detectable effect can still be small. The written-value control uses the same matched pairs for its excess estimate and interval; the exception model’s estimate pools all three demonstration sets and clusters on matched set.

B.2 Naming and written-value controls

Written values. We split generated chains by whether they state the target’s true value. An observation enters the excess estimate only when its neutral twin falls in the same split. The estimate and its 95% interval use those same paired observations. Pooled over eligible cells, the excess is +0.08 points in 132 cells where the value is stated and -0.10 in 27 where it is not. The exception persists in the stated-value split (main text), so this split does not explain its effect.

Identifier-style names. At level 3, four models receive names such as q4 instead of four, under

three demonstration sets. Without CoT, queried-variable excess name errors range from -0.2 to +0.2 points; 0 of 4 cells are reliable. For number words on the same models, the range is -0.2 to +2.8, with 3 of 4 reliable. The numeric name-error effect is therefore stronger for number words in this comparison; congruent identifier names can still improve accuracy.

Word-class effects. With CoT at level 4, a number word on the queried variable costs Llama-3.2-3B 9.9 accuracy points whether the name agrees with the value or not. Those errors restate the wrong equation. This is distinct from answering the digit the name denotes, which is why the name-suggested value result cannot be read as removal of every naming effect.

Unused variables. A number word on the level 4 distractor changes accuracy by at most 1.2 points. Its no-CoT excess name errors range from +2.1 to +2.3; 2 of 2 cells are reliable. With CoT, the range is +0.0 to +1.0, with 0 of 2 reliable.

Mixed-effects check. We compare the paired bootstrap with a mixed logit model, $\text{name_error} \sim \text{incongruent} + C(\text{seed})$, with a random intercept for matched set. Each twin’s outcome refers to the same name-suggested digit. Both procedures use the same fixed subsample of 600 sets per cell. Of 57 fitted cells, 50 agree on whether the interval excludes zero; 11 additional cells have no outcome variation. The mixed fit supports an effect in 5 cells where the bootstrap does not and withdraws support in 2 cells where it does (Gemma-3-12B-it, value-only chain, queried variable; Llama-3.1-Nemotron-Nano-8B, CoT, intermediate variable). Subsampling also removes support from 1 starred Table 3 cell under both procedures (Gemma-3-4B-it, no CoT, queried variable). The mixed model uses variational Bayes, so its interval is an approximate posterior interval. We retain the full-data bootstrap as the primary estimate on the reported percentage-point scale.

Positional-copy control. Across 4110 generated CoT name errors over models, levels and demonstration sets, the final number in the comma-delimited segment before the answer is the name-suggested value in 13% of cases and the target’s true value in 11%. A trailing name-suggested value is therefore absent from many errors. This check constrains the positional-copy explanation

Model	Regime	Neutral accuracy (%)	number word on the <i>queried</i> variable		number word on the <i>intermediate</i> variable	
			helps	errors	helps	errors
Gemma-3-12B-it	CoT	100 [100,100]	+0.0	+0.0	+0.0	+0.0
Gemma-3-12B-it	direct	78 [76,81]	-3.2*	-0.2	-3.7*	+0.4
Gemma-3-4B-it	CoT	100 [100,100]	-0.0	+0.0	-0.0	+0.0
Gemma-3-4B-it	direct	55 [48,59]	+0.2	+0.9*	+0.6	+0.3
Llama-3.1-8B-Instruct	CoT	100 [99,100]	+0.2	+0.0	+0.3	+0.0
Llama-3.1-8B-Instruct	direct	52 [45,65]	+1.7	+0.1	+5.2*	+0.2
Llama-3.1-8B	CoT	95 [86,100]	-1.3	-0.1	+3.8	-0.0
Llama-3.1-8B	direct	53 [41,62]	+4.8*	+1.2*	+3.8*	-0.0
Llama-3.2-3B-Instruct	CoT	100 [100,100]	-0.0	+0.0	-0.2	+0.0
Llama-3.2-3B-Instruct	direct	44 [40,50]	-1.8*	+0.2	+0.5	+0.5
Llama-3.2-3B	CoT	98 [97,100]	-1.2	+0.1	+0.9	-0.2*
Llama-3.2-3B	direct	22 [13,31]	+5.4*	+2.8*	+1.1	+1.8
OLMo-2-7B-Instruct	CoT	95 [93,97]	+0.7*	-0.1	-2.7	-0.0
OLMo-2-7B-Instruct	direct	54 [50,58]	+0.6	-0.2	+3.8*	-0.4

Table 3: Arithmetic level 3. Accuracy (%); name effects (points). *Reliable across demonstration sets.

Model	Regime	Neutral accuracy (%)	number word on the <i>queried</i> variable		number word on the <i>intermediate</i> variable	
			helps	errors	helps	errors
OLMo-2-1B-Instruct	CoT	47 [44,51]	-5.7*	+0.4	+7.0*	+3.2*
OLMo-2-1B-Instruct	direct	34 [30,38]	-0.0	+2.7*	+3.0*	-0.2

Table 4: The level-3 model below the preregistered 90% mean neutral-CoT accuracy floor, excluded from the accuracy-filtered behavioral table. Columns and claim markers follow Table 3.

Contrast	Cells	Equivalent	Reliable
CoT name errors	44	43	3
CoT accuracy	88	61	15
Direct name errors	44	31	13

Table 5: Equivalence within the two-point margin uses 90% intervals; reliability uses 95% intervals excluding zero and the same sign across all three demonstration sets. Each accuracy cell is one facilitation or interference contrast.

of Liu (2026), but does not identify what was copied in each case.

B.3 Derived checkpoints

We compare six checkpoints on the Llama-3.1-8B architecture: pretrained, instruction-tuned, single-task LoRA, multi-task LoRA, TIES-merged adapters and reasoning-tuned. Adapters are folded into the weights before evaluation. Shared tokeniz-

ers and identical matched problems permit paired comparisons (Table 7). A pretrained/instruct Gemma pair supplies a second family comparison.

Task accuracy complicates interpretation. The multi-task model’s reliable negative excess (-0.58) accompanies lower neutral accuracy (46.1%, versus 52.2%). The reasoning-tuned checkpoint has the lowest accuracy (39.4%). It also requires a system instruction to enable long-form reasoning; our raw-completion prompts leave that mode off. The result concerns the checkpoint under this prompt, not reasoning mode in deployment. Across these checkpoints, CoT excess name errors are at most 0.20 points in magnitude.

B.4 Values without equations

Our chain restates each equation before substituting into it, as Kudo et al.’s does. Their Tables 5–6 also report a format that writes only the sub-results, cup=5, pen=6. We test this format at

Model	Level	Regime	Neutral accuracy (%)	number word on the <i>queried</i> variable		number word on the <i>intermediate</i> variable	
				helps	errors	helps	errors
Llama-3.1-8B	1	CoT	99 [98,100]	+0.0	+0.0	+0.3	+0.0
Llama-3.1-8B	1	direct	74 [38,99]	+16.1*	+11.2*	+8.8*	+1.3
Llama-3.2-3B	1	CoT	97 [95,100]	+1.1	+0.1	-2.4	+0.3
Llama-3.2-3B	1	direct	46 [21,93]	+0.6	+5.9*	-3.9*	+2.6*
Llama-3.1-8B	2	CoT	100 [100,100]	-0.1	+0.0	-0.0	+0.0
Llama-3.1-8B	2	direct	36 [33,37]	+11.8*	+0.7	+2.6	+2.2
Llama-3.2-3B	2	CoT	94 [90,98]	+3.7	+0.5*	+1.0*	-0.1
Llama-3.2-3B	2	direct	29 [27,30]	+10.8*	+2.5*	+2.6*	+2.2
Llama-3.1-8B	4	CoT	88 [63,100]	+1.0	-0.0	-2.3	+0.2
Llama-3.1-8B	4	direct	28 [26,31]	+13.7*	+5.8	+6.0*	+0.2
Llama-3.2-3B	4	CoT	64 [26,85]	-10.2*	+0.7	-2.6*	+0.4
Llama-3.2-3B	4	direct	20 [16,23]	-1.6	+1.9	-0.2	+1.2*
Llama-3.1-8B	5	CoT	65 [43,100]	+1.2	-0.3	+0.1	-0.1
Llama-3.1-8B	5	direct	20 [13,26]	+20.2*	+20.5*	+0.9	+0.4
Llama-3.2-3B	5	CoT	69 [48,97]	-1.4	+0.1	-2.5	-0.1
Llama-3.2-3B	5	direct	12 [11,12]	+4.7*	+3.5*	+1.1*	+0.5

Table 6: Additional task levels for the two Llama models. Columns and claim markers follow Table 3; level 1 states the intermediate value directly.

level 3 with three demonstration sets. Two predictions were registered for both Llama models: that the excess name errors at the intermediate’s value step would exceed the 2-point margin, and that the value would still be decodable there (probe accuracy at least 0.85), making any failure one of readout.

Both predictions fail as registered. Llama-3.1-8B writes the name-suggested value at the intermediate’s value step +0.15 points above its twin, inside the margin. Llama-3.2-3B is at +9.20, which passes our claim rule, but across the three demonstration sets that is +1.7 / +24.2 / +1.7, so one set carries most of it. More informative is what the probes say. At the step that writes the value, the true value is decodable at 0.81 and 0.59 for Llama-3.1-8B and at 0.61 and 0.79 for Llama-3.2-3B, against 1.00 and 0.99 for the same model under the full chain. Neutral accuracy falls with it, to 54% and 31%.

Removing the equations lowers both accuracy and true-value probe readouts. It therefore fails to isolate a readout error while preserving those measures.

C Probe and patching controls

Probes predict the true digit, the name-suggested digit and their probability margin. The name-identity control assigns a fixed random digit to each name; selectivity subtracts its accuracy from true-value accuracy. Layers in the supplied-calculation analysis are selected by neutral selectivity, averaged over three probe seeds. P1 is the name token, P2 the end of its defining equation, P3 the query, P4 the position before its value is written, and P5 the position before the final answer. P1 is a reference, and P3 can reveal name identity rather than a computed value. Figure 6 gives the token and layer sweep supporting the main readout analysis.

Patches replace residual-stream activations at matched name-token positions, at individual layers or across all layers. The neutral-twin source tests removal of the name error; a different ordinary name tests damage to correct answers; an alternative misleading name tests specificity. The span sweep instead replaces an equation, the query or the value-writing segment in four-layer windows. Figure 7 supports the early-layer lo-

Tuning	Model	Queried	Intermediate	Acc.
base	Llama-3.1-8B	+1.20*	-0.02	53.4
instruct	+ instruct	+0.15	+0.22	52.2
single-task finetune	+ task LoRA	+0.05	+0.23	52.2
multi-task finetune	+ multi-task LoRA	-0.58*	-0.08	46.1
weight merge	+ TIES merge	-0.10	+0.10	51.9
reasoning-tuned	Nemotron-Nano-8B	+1.35*	+0.33	39.4
base	Gemma-3-4B	+0.08	+0.02	60.2
instruct	+ instruct	+0.88*	+0.33	54.9

Table 7: Level 3, no CoT, name errors above the matched twin’s rate, mean over three demonstration sets. *Acc.* is neutral accuracy. * as in Table 3. The bottom block repeats the pretrained-to-instruct step in a second family, Gemma-3-4B.

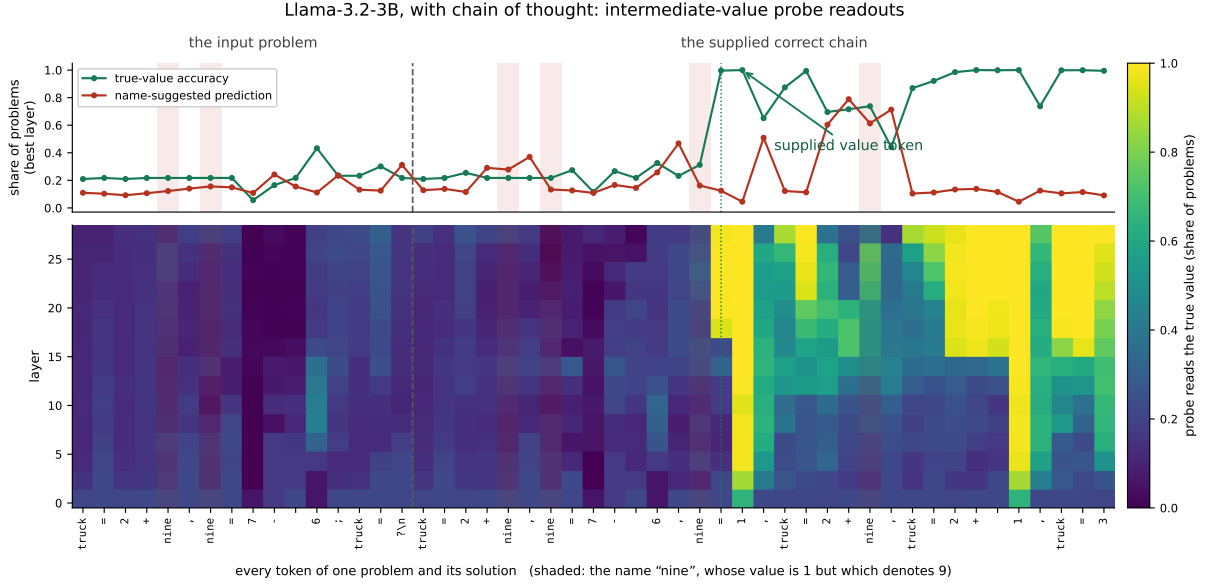


Figure 6: Llama-3.2-3B intermediate-value readouts under supplied correct calculations. Rates pool problems; tokens illustrate one example.

calization. Table 8 gives the answer-only control rates and denominators for both Llama models. OLMo’s supplied-calculation controls are reported in Table 10. We also reverse the main contrast, transplanting misleading-name states into the ordinary-name twin. OLMo’s largest injection rate across layers and both variables is 2.7% at layer 5; word-control damage there is 0.4%. The injection itself damages 7.0% of correct answers. This selected maximum is descriptive.

Instance-level association. Within each model, level, regime, variable and position, we regress name errors on the probe margin, $\log p_{\text{true}} - \log p_{\text{name}}$, where p_{name} is the probe probability of the name-suggested value, with standard errors clustered on matched set. The layer is selected by neutral accuracy. Of 63 fits, 29 have $p < .05$: 24 coefficients are negative and 5 positive. The

direction is not consistent, and 4 positive results come from the model supplying most fits. This leaves the instance-level association unresolved. The model-level pattern is also sensitive to panel composition (Appendix D).

C.1 Probe recipe and patching scope

We check two implementation choices inherited from Kudo et al. (2026): the probe optimizer and the name occurrences included in a patch.

Recipe. We refit the level-3 probes for Llama-3.2-3B with L-BFGS, and with SGD on standardized inputs, against the reported recipe.

All three recipes give the same qualitative result: the value is decodable where the chain writes it (0.96–1.00) and not at the end of its defining equation (0.18–0.50). The query position moves, from 0.271 under the reported recipe to 0.393 un-

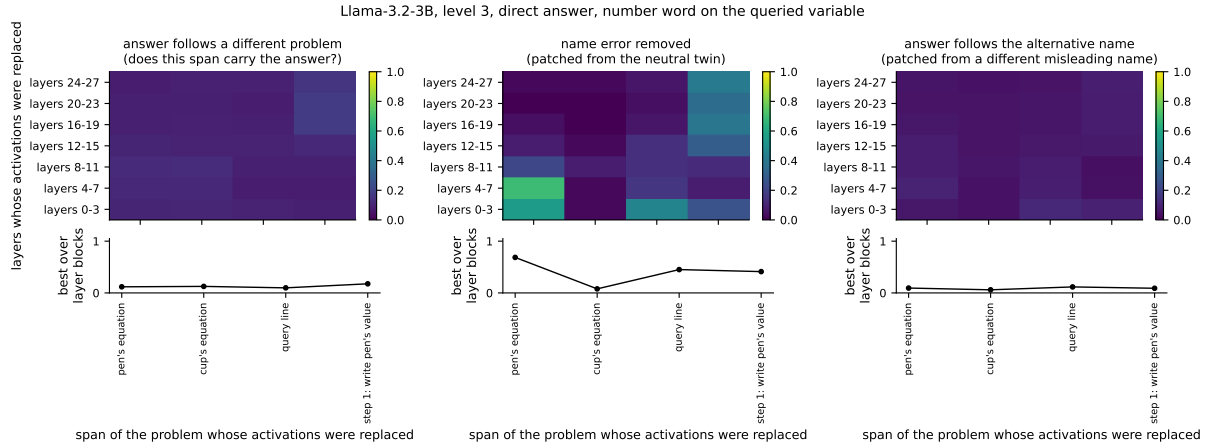


Figure 7: Llama-3.2-3B answer-only patches in four-layer windows. Bottom: maximum over windows.

Model	Variable	Pairs	Name errors	Correct	Neutral	Alternative	Word damage
Llama-3.2-3B	queried	2000	172	272	51.7	34.9	22.4
Llama-3.1-8B	intermediate	2000	72	838	30.6	30.6	8.5

Table 8: All-layer answer-only patches, demonstration set 7. Neutral and alternative: percentage of baseline name errors removed. Word damage: percentage of correct ordinary-name answers changed; *Correct* gives that denominator.

der standardization, within a range of 0.18–0.39 over recipes and variables. We already report that position as the presence of the name rather than a decoded value, because its control task scores 0.97, so this variation does not change the interpretation of that position.

Scope. Patching every occurrence of the name includes the copy the model writes in its own output. In the direct regime, restricting the patch to the name’s occurrences in the prompt leaves the intermediate variable essentially unchanged, since its name never appears in the answer: removal is 43% at layers 0–3 under both scopes. For the queried variable the best windows remove similar shares (52% at layers 4–7 prompt-only, against 53% at layers 0–3), the prompt-only patch removes less in the later windows, and the two disagree most in the first window, where the prompt-only patch removes 3% and damages 76% of correct answers. That is mechanical: the queried variable’s name also sits at the position we read, so patching its prompt copies alone leaves the model reading a name whose early representation no longer matches. The localization we report does not depend on the choice.

D Task competence and value readouts

Table 11 compares both variables for 9 models. Layers are selected by neutral selectivity, averaging three probe seeds. Task accuracy and paired excess name errors pool demonstration sets 7, 11 and 13; probes use set 7. The within-model difference holds overall task accuracy fixed but does not isolate a causal effect of the readout. We average the two variable readouts within each model before computing descriptive correlations. Table 12 repeats them after omitting each model. Roles are not independent observations. With this small, heterogeneous panel, correlations cannot distinguish overall competence from the role of decodable information.

E Gold-trained transfer to generated chains

We read the state before the first numeric assignment to the target in each saved generated chain. Expression operands are not counted as completed value writes; tokens that include any part of the supplied value are excluded. A fixed sample of 200 matched sets is selected by a hash independent of outcomes, shared across models, variables and demonstration sets. All additional chains with name errors and their neutral twins are evaluated

Position	Variable	SGD (reported)	L-BFGS	standardized
defining equation	queried	0.184	0.190	0.183
defining equation	intermediate	0.496	0.422	0.420
query	queried	0.183	0.316	0.280
query	intermediate	0.271	0.366	0.393
value step	queried	1.000	1.000	1.000
value step	intermediate	0.982	0.962	0.972
answer	queried	1.000	1.000	1.000

Table 9: Best-layer true-value accuracy on neutral problems under three probe recipes (proportions from 0 to 1).

Target	Read	Errors	Neutral removed	Word dmg.	Alt. removed	Alt. followed
v1	value	0	—	1.6	—	0.0
v1	answer	0	—	1.6	—	0.0
v2	value	28	92.9	7.3	82.1	4.0
v2	answer	3	100.0	0.8	100.0	0.6

Table 10: OLMo-2-1B-Instruct, supplied correct calculations, 500 matched pairs per target. All-layer rates (%): neutral/alternative name-error removal, word-control damage to correct predictions, and predictions following the alternative digit. Roles v1/v2: queried/intermediate; —: undefined.

separately. Neutral gold-chain probes use 10,000 separate training problems, three probe seeds and the original neutral-selectivity layer. Table 13 tests transfer to held-out neutral generated chains. A high-accuracy interpretation requires at least 90% neutral calibration in that demonstration set. Cases below that threshold yield descriptive readouts only. The augmentation is selected by observed errors and cannot estimate their population rate. Appendix F separately trains probes on generated neutral chains and tests the same held-out prefixes. Llama-3.2-3B meets the neutral transfer threshold in 6 of 6 cells, with calibration accuracy from 97.2 to 100.0%. OLMo-2-1B meets the neutral transfer threshold in 0 of 6 cells, with calibration accuracy from 19.4 to 62.2%. Name-error cases in the competent model are few; error-conditioned readouts cannot establish a general mechanism from that subset. Failed neutral calibration prevents a strong generated-chain interpretation for the exception model.

F Training probes on generated neutral chains

Gold-trained probe transfer (Appendix E) may fail because the states along generated chains differ from those along the supplied correct chain. We therefore train on freely generated neutral chains from 2,000 source problems and select layers on 400 separate validation problems. Canonical equations partition the source pool, keeping renamed

versions of a computation together. The training and validation pools contain 630 and 155 computations; both exclude all test and demonstration computations. These counts distinguish independent computations from repeated name renderings. Generation is greedy with demonstration set 7 and a 128-token limit. The probe sees the hidden state before the target variable’s first numeric assignment, with every token boundary checked to exclude the value digit. Missing or unsafe boundaries are excluded and counted. We retain all usable neutral chains regardless of whether the model writes the correct value, using the executed true value as the training label. Three probe seeds use the original full-batch SGD recipe (learning rate 0.001, 10,000 epochs). States are stored in float32 and nonfinite states or fitted weights are rejected. Layers are searched at stride two, including the final layer, and chosen only by mean validation accuracy with the lowest-layer tie break. The name-identity control is trained separately and does not select layers. Table 14 reports the usable source counts and selected layers. Evaluation uses exactly the earlier fixed 200-set sample and separately reported additional name-error cases under demonstration sets 7, 11 and 13. Table 15 gives held-out neutral calibration and boundary coverage. Intervals use 4,000 canonical-computation bootstrap draws, keeping repeated names together. The 90% calibration threshold applies to the point estimate, not the lower interval

Model	Role	Layer	Task	Neutral	Name ID	Misleading	Excess
Gemma-3-12B-I	v1	40	100.0	100.0	13.1	100.0	0.0
Gemma-3-12B-I	v2	42	100.0	100.0	11.9	100.0	0.0
Gemma-3-4B	v1	25	100.0	100.0	10.0	100.0	0.0
Gemma-3-4B	v2	28	100.0	99.8	12.4	100.0	0.0
Gemma-3-4B-I	v1	29	100.0	100.0	7.4	99.9	0.0
Gemma-3-4B-I	v2	28	100.0	99.8	10.0	99.9	0.0
Llama-3.1-8B	v1	20	95.2	100.0	20.5	100.0	-0.1
Llama-3.1-8B	v2	22	95.2	99.4	24.9	99.2	-0.0
Llama-3.1-8B-I	v1	20	99.7	100.0	21.0	100.0	0.0
Llama-3.1-8B-I	v2	20	99.7	100.0	25.7	100.0	0.0
Llama-3.2-3B	v1	19	98.3	99.9	22.7	100.0	0.1
Llama-3.2-3B	v2	21	98.3	97.6	31.5	99.6	-0.2
Llama-3.2-3B-I	v1	18	100.0	99.9	26.3	100.0	0.0
Llama-3.2-3B-I	v2	19	100.0	99.9	29.5	99.9	0.0
OLMo-2-1B	v1	14	47.2	99.7	48.9	100.0	0.4
OLMo-2-1B	v2	0	47.2	21.8	15.0	21.8	3.2
OLMo-2-7B	v1	22	95.3	100.0	37.7	100.0	-0.1
OLMo-2-7B	v2	26	95.3	99.4	47.5	97.6	-0.0

Table 11: Task and true-digit probe accuracies (%), name-identity control accuracy (%), and excess name errors (points). Roles v1/v2 are queried/intermediate. Selectivity is neutral minus name-ID accuracy. Below-floor models remain descriptive.

Omitted model	Task-excess	Probe-excess	Task-probe
All models	-0.99	-1.00	+1.00
Gemma-3-12B-I	-0.99	-1.00	+1.00
Gemma-3-4B	-0.99	-1.00	+1.00
Gemma-3-4B-I	-0.99	-1.00	+1.00
Llama-3.1-8B	-0.99	-1.00	+1.00
Llama-3.1-8B-I	-0.99	-1.00	+1.00
Llama-3.2-3B	-0.99	-1.00	+1.00
Llama-3.2-3B-I	-0.99	-1.00	+1.00
OLMo-2-1B	+0.98	+0.78	+0.85
OLMo-2-7B	-0.99	-1.00	+1.00

Table 12: Descriptive Pearson correlations of model means; Spearman values and within-model variable differences are in the saved summary. No significance or causal claim is based on these coefficients.

endpoint. Llama passes all six calibration cells; OLMo passes none. Table 16 retains error readouts with their denominators. OLMo’s readouts remain descriptive. Llama has only one name-error case in the primary sample and thirteen additional observed writes; these rare, selected cases cannot establish a general mechanism.

G Paired change from answer-only to written calculations

We compare prompting conditions on exactly the same matched sets and demonstration seeds used in the 44 name-error equivalence cells. Each observation subtracts the ordinary-name rate from the misleading-name rate, then subtracts the answer-only contrast from the CoT contrast. Negative changes indicate fewer added name-suggested answers. We bootstrap matched sets, keeping all three demonstration repeats together (4,000

draws). Reliable reductions require a negative change in every demonstration set and a 95% interval below zero. The panel is descriptive across related checkpoints; its cells are not independent model replications. Of the 44 paired changes, 10 meet that rule; 19 have intervals below zero. A sensitivity bootstrap that also clusters repeated arithmetic computations yields 10 reliable reductions. Intervals are pointwise; the count is not a multiplicity-corrected family-wide test. Before writing calculations, 31 cells already meet the two-point equivalence bound, compared with 43 afterward. Thus the equivalence count is not a count of significant improvements. Accuracy gains accompany many reductions and do not identify an effect independent of arithmetic competence.

Model	Set	Role	Neutral coverage	Calibration (%)	Sample errors	Extra errors
OLMo-2-1B	7	v1	194/200	62.2 [55.2, 68.7]	8	69
OLMo-2-1B	7	v2	186/200	19.4 [14.0, 25.3]	11	82
OLMo-2-1B	11	v1	187/200	55.3 [48.3, 62.6]	7	59
OLMo-2-1B	11	v2	129/200	21.7 [14.7, 28.7]	2	21
OLMo-2-1B	13	v1	193/200	52.7 [45.6, 59.8]	17	159
OLMo-2-1B	13	v2	124/200	24.2 [16.9, 32.3]	1	30
Llama-3.2-3B	7	v1	200/200	98.0 [96.0, 99.5]	1	8
Llama-3.2-3B	7	v2	200/200	97.2 [94.8, 99.0]	0	0
Llama-3.2-3B	11	v1	199/200	100.0 [100.0, 100.0]	0	0
Llama-3.2-3B	11	v2	200/200	100.0 [100.0, 100.0]	0	0
Llama-3.2-3B	13	v1	199/200	99.5 [98.5, 100.0]	0	5
Llama-3.2-3B	13	v2	200/200	99.5 [98.5, 100.0]	0	0

Table 13: Gold-trained probe accuracy on neutral generated chains [95% interval], and eligible name-error counts. Extra cases exclude the fixed sample. Roles v1/v2 are queried/intermediate. Calibration is required before interpreting error readouts.

Model	Role	Train n	Validation n	Layer	Validation (%)	Name ID (%)
OLMo-2-1B-I	v1	1924	390	14	47.9	41.6
OLMo-2-1B-I	v2	1877	376	16	87.8	86.6
Llama-3.2-3B	v1	1994	398	22	97.5	26.8
Llama-3.2-3B	v2	2000	400	20	99.2	28.6

Table 14: Usable generated-boundary counts and validation-selected layers. Roles v1/v2 are queried/intermediate. Source cohorts are computation-disjoint; validation scores are not test estimates.

H Response formats and fixed correct calculations

These exploratory controls use Llama-3.2-3B, Llama-3.1-8B and OLMo-2-1B-Instruct, level 3 and demonstration seeds 7, 11 and 13. The protocol and cohorts were frozen before collecting outputs. Intervals use 4,000 bootstrap draws over arithmetic computations, keeping repeated names and demonstrations together. Intervals are point-wise, without multiplicity correction; zero-event bootstrap intervals do not establish zero population rates.

Equations and repeated names. We cross numeric equations with name repetition in demonstrated outputs. For example, named equations write $\text{cup}=2+3=5$, $\text{pen}=1+5=6$; unnamed equations write $2+3=5$, $1+5=6$; values-only formats write $\text{cup}=5$, $\text{pen}=6$ or 5 , 6 . All end with the same Answer: 6 . Only examples provide correct calculations; test outputs are freely generated. Repeated ordinary-word notes before each example output match full prompt token counts across formats and matched names. Generated lengths are measured rather than forced to match. We select 200 independent test computations, each rendered with ordinary names and with a misleading name on either variable, yielding 600 paired ob-

servations per target across demonstrations. Calibration uses 100 independent ordinary-name computations, disjoint from tests and demonstrations. We try 3, 8 and 16 examples, seeking at least 90% accuracy in every format and a range within two points. No model passes; all use 16 examples. A held-out comparison matches competence only if both ordinary-name accuracies reach 90% and their paired 90% difference interval fits within ± 2 points. Only the named-versus-unnamed equation comparisons in the two Llama models pass (four target comparisons). None of the equation-versus-values comparisons passes.

Answering after a correct calculation. We resume after actual original generations for which every restatement, substitution and value matches the executable gold calculation. Both naming twins must qualify. Within each naming condition, the input and complete generated calculation are fixed. We cross a final query repeating the queried name with the original requested value=?, and an output starting with Answer: with one starting with Answer: $\text{pen}=?$. The instruction always requests the original problem's value. These continuations are conditional on successful original calculations; they do not estimate an effect of making a calculation correct. Tail lengths and output markers differ by design. Each Llama model sup-

Model	Role	Set	Neutral coverage	Calibration (%)	Name ID (%)	Pass
OLMo-2-1B-I	v1	7	194/200	61.3 [54.1, 68.6]	37.3 [31.2, 43.4]	no
OLMo-2-1B-I	v1	11	187/200	52.8 [45.1, 60.3]	32.4 [26.6, 38.5]	no
OLMo-2-1B-I	v1	13	193/200	50.3 [43.0, 57.4]	35.2 [29.1, 41.6]	no
OLMo-2-1B-I	v2	7	186/200	78.0 [71.5, 83.9]	81.9 [76.9, 86.7]	no
OLMo-2-1B-I	v2	11	129/200	80.1 [72.8, 86.8]	56.6 [49.6, 63.3]	no
OLMo-2-1B-I	v2	13	124/200	71.8 [64.2, 78.4]	46.8 [37.8, 55.6]	no
Llama-3.2-3B	v1	7	200/200	98.5 [96.7, 100.0]	18.5 [13.8, 23.7]	yes
Llama-3.2-3B	v1	11	199/200	100.0 [100.0, 100.0]	13.9 [9.5, 18.5]	yes
Llama-3.2-3B	v1	13	199/200	97.2 [94.2, 99.5]	16.9 [12.3, 21.8]	yes
Llama-3.2-3B	v2	7	200/200	96.3 [93.6, 98.6]	28.8 [22.7, 35.0]	yes
Llama-3.2-3B	v2	11	200/200	100.0 [100.0, 100.0]	17.2 [12.4, 22.1]	yes
Llama-3.2-3B	v2	13	200/200	97.7 [95.5, 99.3]	17.8 [13.3, 22.8]	yes

Table 15: Held-out primary neutral generated chains. Coverage counts valid pre-value boundaries over all primary neutral rows. Accuracy intervals are 95% canonical-computation bootstrap intervals; pass means the calibration point estimate meets 90%.

Model	Role	Set	Sample n	Sample accuracy	Extra n	Extra accuracy
OLMo-2-1B-I	v1	7	8	4.2 [0.0, 12.5]	69	2.9 [0.0, 7.0]
OLMo-2-1B-I	v1	11	7	19.0 [0.0, 38.1]	59	10.7 [4.2, 18.8]
OLMo-2-1B-I	v1	13	17	21.6 [7.8, 39.2]	159	13.6 [8.6, 19.4]
OLMo-2-1B-I	v2	7	11	33.3 [15.2, 51.5]	82	18.7 [11.4, 27.2]
OLMo-2-1B-I	v2	11	2	0.0 [0.0, 0.0]	21	17.5 [3.3, 31.8]
OLMo-2-1B-I	v2	13	1	0.0 [0.0, 0.0]	30	25.6 [13.7, 39.1]
Llama-3.2-3B	v1	7	1	0.0 [0.0, 0.0]	8	12.5 [0.0, 42.9]
Llama-3.2-3B	v1	11	0	—	0	—
Llama-3.2-3B	v1	13	0	—	5	0.0 [0.0, 0.0]
Llama-3.2-3B	v2	7	0	—	0	—
Llama-3.2-3B	v2	11	0	—	0	—
Llama-3.2-3B	v2	13	0	—	0	—

Table 16: True-digit accuracy (%) [95% interval] on actual name errors, with the fixed sample and additional error-selected cases separated. Extra cases exclude the primary sample. Dashes denote empty strata. OLMo fails neutral calibration, so its readouts are descriptive.

plies 200 pairs per demonstration set and target. OLMo supplies 144/108/116 queried-target pairs and 44/8/25 intermediate-target pairs. Across demonstration sets, the queried-target cohorts contain 210/234/205 distinct computations for Llama-3B/Llama-8B/OLMo; intermediate-target cohorts contain 218/233/53. Generation budgets are 128 tokens for format tests and 16 for final continuations, with greedy full-vocabulary decoding. Unparsed outputs count as incorrect. For OLMo, repeating the queried name lowers added errors by 12.2 points [8.4, 16.1] in digit outputs, but ordinary-name accuracy rises from 78.0% to 96.2%. An assignment output also changes accuracy. No OLMo final-format comparison meets the competence criterion. With the misleading name on the intermediate variable, all added-error estimates are zero; OLMo has one suggested-digit answer in each naming condition in the repeated-query/digit format. The query repeats an ordinary name in this control. Neither Llama produces a

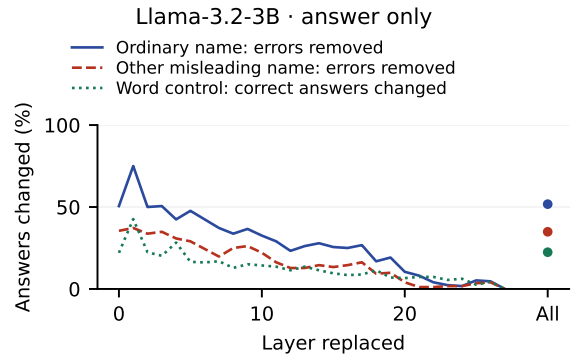


Figure 8: Arithmetic name-site patches, Llama-3.2-3B, queried variable, set 7. All replaces every layer.

suggested-digit answer in any final format. These observations distinguish correct calculations from reliable final answering, without identifying a single responsible feature.

Model	Level/role	Answer-only	CoT	Paired change [95% interval]	Accuracy gain
Llama-3.2-3B	L1/v1	+5.90	+0.12	-5.78 [-6.55, -5.02]*	+50.7
Llama-3.2-3B	L1/v2	+2.62	+0.30	-2.32 [-2.87, -1.77]*	+50.7
Llama-3.1-8B	L1/v1	+11.20	+0.00	-11.20 [-12.05, -10.37]*	+25.6
Llama-3.1-8B	L1/v2	+1.33	+0.02	-1.32 [-1.68, -0.97]	+25.6
Llama-3.2-3B	L2/v1	+2.50	+0.53	-1.97 [-2.65, -1.28]*	+65.1
Llama-3.2-3B	L2/v2	+2.20	-0.05	-2.25 [-2.92, -1.63]	+65.1
Llama-3.1-8B	L2/v1	+0.73	+0.02	-0.72 [-1.25, -0.20]	+64.4
Llama-3.1-8B	L2/v2	+2.20	+0.00	-2.20 [-2.73, -1.63]	+64.4
Llama-3.2-3B	L3/v1	+2.78	+0.13	-2.65 [-3.30, -2.00]*	+76.1
Llama-3.2-3B	L3/v2	+1.78	-0.17	-1.95 [-2.42, -1.48]	+76.1
Llama-3.2-3B-I	L3/v1	+0.17	+0.00	-0.17 [-0.62, +0.28]	+56.0
Llama-3.2-3B-I	L3/v2	+0.47	+0.00	-0.47 [-1.05, +0.12]	+56.0
Llama-3.1-8B	L3/v1	+1.20	-0.13	-1.33 [-1.82, -0.85]*	+41.8
Llama-3.1-8B	L3/v2	-0.02	-0.02	+0.00 [-0.47, +0.47]	+41.8
Llama-3.1-8B-I	L3/v1	+0.15	+0.00	-0.15 [-0.60, +0.28]	+47.5
Llama-3.1-8B-I	L3/v2	+0.22	+0.02	-0.20 [-0.68, +0.28]	+47.5
Gemma-3-4B-I	L3/v1	+0.88	+0.00	-0.88 [-1.38, -0.38]*	+45.1
Gemma-3-4B-I	L3/v2	+0.33	+0.02	-0.32 [-0.80, +0.15]	+45.1
Gemma-3-12B-I	L3/v1	-0.23	+0.00	+0.23 [-0.08, +0.55]	+22.4
Gemma-3-12B-I	L3/v2	+0.40	+0.00	-0.40 [-0.83, +0.02]	+22.4
OLMo-2-1B-I	L3/v1	+2.73	+0.45	-2.28 [-3.28, -1.30]*	+13.5
OLMo-2-1B-I	L3/v2	-0.20	+3.17	+3.37 [+2.37, +4.37]	+13.5

Table 17: Excess name errors and paired CoT-minus-answer-only changes (percentage points). Accuracy gain is the ordinary-name accuracy change. Roles v1/v2/v3 are the first/second/third variables in the problem; the queried role is v2 at level 2 and v1 otherwise. *Reliable reduction across all demonstration sets.

I Code task levels and identifiers

Levels vary dependency depth, distractors and which assignment receives the misleading name. The primary comparison uses level 5. A function is generated once; renaming preserves its statements, input and executable result. Table 22 lists the operation suggested by each identifier. Sum and length are admissible name-suggested operations on the aggregate step; minimum and maximum are excluded because their result is already present in the input.

J Identifier meanings

For each identifier and model, we compare log-probabilities of candidate right-hand sides after a prefix such as `def f(xs):\n total =`. The highest-scoring operation records what that identifier evokes. Of 25 names, 12 match the intended operation in at least 80% of models. Most sum-family names instead evoke length in 2 of seven models. This check matters when interpreting a small effect: an identifier need not suggest the same operation to every model. The design planned to retain only names passing the 80% panel gate. The runs instead retained the full name table, so the reported comparisons depart from that selection rule. The results concern these named groups; they do not establish that every model

reads each name as intended.

K Operation and name pairings

Table 24 reports all tested aggregate/name pairings. The reverse of the main length-to-sum conflict is weak at level 5: sum-computed variables with length-like names have excess within two points for models above the neutral-accuracy floor. At level 3, OLMo-2-7B has +4.4 points of excess in that pairing. Gemma-3-4B shows no main-cell effect, consistent with its identifier-meaning check. OLMo-2-1B’s neutral accuracy is only 37.8%, making its small, broadly distributed effects harder to interpret as a specific trace failure.

L Demonstration formats

All few-shot comparisons hold programs, demonstration identities and model weights fixed. The values-only format writes `qq = 3`; the expression-and-value format writes `qq = len(xs) = 3`; the interactive Python format writes `>>> qq` followed by `3`; comment mode annotates each assignment with its value. These are model-generated transcripts, not interpreter executions. Prose asks for step-by-step reasoning without demonstrations. Table 26 gives the paired excess and interval for each few-shot format.

On the main conflict, interactive-transcript excess is +9.5 and +12.6 points for OLMo-2-7B and

Model	Level/role	Answer-only	CoT	Paired change [95% interval]	Accuracy gain
OLMo-2-7B-I	L3/v1	-0.23	-0.08	+0.15 [-0.33, +0.60]	+41.0
OLMo-2-7B-I	L3/v2	-0.42	-0.03	+0.38 [-0.13, +0.87]	+41.0
Llama-8B single-task	L3/v1	+0.05	+0.00	-0.05 [-0.50, +0.42]	+47.6
Llama-8B single-task	L3/v2	+0.23	-0.02	-0.25 [-0.75, +0.23]	+47.6
Llama-8B multi-task	L3/v1	-0.58	+0.00	+0.58 [+0.23, +0.95]	+53.7
Llama-8B multi-task	L3/v2	-0.08	+0.00	+0.08 [-0.38, +0.53]	+53.7
Llama-8B merged	L3/v1	-0.10	+0.00	+0.10 [-0.30, +0.50]	+47.8
Llama-8B merged	L3/v2	+0.10	+0.00	-0.10 [-0.58, +0.38]	+47.8
Nemotron-Nano-8B	L3/v1	+1.35	+0.08	-1.27 [-1.78, -0.77]	+58.8
Nemotron-Nano-8B	L3/v2	+0.33	+0.20	-0.13 [-0.63, +0.38]	+58.8
Gemma-3-4B	L3/v1	+0.08	+0.00	-0.08 [-0.53, +0.35]	+39.8
Gemma-3-4B	L3/v2	+0.02	+0.00	-0.02 [-0.52, +0.48]	+39.8
Llama-3.2-3B	L4/v1	+1.93	+0.68	-1.25 [-1.78, -0.70]	+44.4
Llama-3.2-3B	L4/v2	+1.17	+0.37	-0.80 [-1.32, -0.27]	+44.4
Llama-3.1-8B	L4/v1	+5.83	-0.03	-5.87 [-6.62, -5.17]	+59.5
Llama-3.1-8B	L4/v2	+0.23	+0.20	-0.03 [-0.45, +0.37]	+59.5
Llama-3.2-3B	L5/v1	+3.45	+0.08	-3.37 [-4.10, -2.67]*	+57.5
Llama-3.2-3B	L5/v2	+0.52	-0.07	-0.58 [-1.17, +0.03]	+57.5
Llama-3.2-3B	L5/v3	-0.35	-0.08	+0.27 [-0.35, +0.88]	+57.5
Llama-3.1-8B	L5/v1	+20.50	-0.30	-20.80 [-21.85, -19.75]*	+45.1
Llama-3.1-8B	L5/v2	+0.45	-0.15	-0.60 [-1.28, +0.05]	+45.1
Llama-3.1-8B	L5/v3	+0.33	+0.08	-0.25 [-0.92, +0.42]	+45.1

Table 18: Excess name errors and paired CoT-minus-answer-only changes (percentage points). Accuracy gain is the ordinary-name accuracy change. Roles v1/v2/v3 are the first/second/third variables in the problem; the queried role is v2 at level 2 and v1 otherwise. *Reliable reduction across all demonstration sets.

Llama-3.2-3B, compared with +0.2 for Llama-3.1-8B. Comments leave +6.8 points in Llama-3.2-3B and near-zero excess in the other two. Prose has little initial name errors, so asking it to state or recompute values does not test removal of the large trace effect. The prediction that interactive transcripts would stay within a factor of two of trace was not supported; the comment prediction held only for two models.

The paired trace-minus-trace-expr name-error-rate reductions are 55 points [49.9, 60.2], 43 [38.2, 47.7] and 12 [8.8, 16.0] for OLMo-2-7B, Llama-3.2-3B and Llama-3.1-8B. These 95% intervals keep each matched program’s demonstration-seed repeats together.

M Probe and patching details

We follow Kudo et al.’s 10-class linear-probe recipe, with three seeds and neutral-only training. Each probe predicts a target digit from one residual-stream position and layer. Name-identity controls use deterministic random labels keyed on the identifier (Hewitt and Liang, 2019). Neutral names and values vary independently, and misleading names are absent from training. The original implementation reused test programs in training. The reported reruns replace those scores: they train on 2,000 independent neutral programs and evaluate 285 misleading-name programs, with

overlap guards that ignore name spelling. Table 27 reports the name controls alongside digit accuracy.

Figure 9’s probe curves use demonstration set 7. Repeating training and evaluation with sets 11 and 13 gives true-digit accuracies of 95–97%, 94–100% and 100–100% across the three sets. Each repeat uses the same independent training programs and three probe seeds; its layer is selected by that set’s neutral accuracy. These repeats assess prompt sensitivity, not a causal role for the decoded digit. Table 27 gives both conditions’ readouts and controls.

We supply the correct trace token by token when caching probe inputs. The decision position is the = before the first traced value; later values cannot affect this position under causal attention. Patching uses only the prompt and this trace prefix. Name patches replace every aligned occurrence of the identifier; decision patches replace only that writing position. Both use one layer at a time and include a neutral-for-neutral disruption control (Zhang and Nanda, 2024). The alternative-name control is uninformative in the main cell: every sum-family identifier implies the same sum, so there is no distinct source name-suggested value. Llama-3.1-8B’s affected two-token name lacks aligned neutral twins for the name-site intervention.

Model	Example output	Correct answers (%)			Added name errors [95% interval]	
		Ordinary	Queried	Intermediate	Queried	Intermediate
Llama-3.2-3B	Equations + names	100.0	100.0	100.0	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
Llama-3.2-3B	Equations only	99.0	99.0	99.5	+0.0 [+0.0, +0.0]	-0.2 [-0.5, +0.0]
Llama-3.2-3B	Values + names	66.3	67.0	68.5	+0.0 [-1.3, +1.3]	+0.5 [-1.0, +2.0]
Llama-3.2-3B	Values only	22.3	20.8	21.5	+0.8 [-1.2, +2.8]	+1.2 [-0.8, +3.2]
Llama-3.1-8B	Equations + names	100.0	100.0	100.0	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
Llama-3.1-8B	Equations only	100.0	100.0	100.0	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
Llama-3.1-8B	Values + names	84.5	82.5	86.8	-0.3 [-1.3, +0.5]	-0.8 [-2.0, +0.0]
Llama-3.1-8B	Values only	38.0	36.5	36.0	-0.5 [-2.2, +1.0]	+0.2 [-1.2, +1.5]
OLMo-2-1B-I	Equations + names	99.7	94.7	93.5	+0.2 [+0.0, +0.5]	+0.3 [+0.0, +0.8]
OLMo-2-1B-I	Equations only	67.3	44.3	53.5	+0.2 [-0.8, +1.2]	+0.0 [-1.2, +1.2]
OLMo-2-1B-I	Values + names	24.7	25.7	25.2	-0.2 [-2.8, +2.5]	-1.3 [-3.8, +1.2]
OLMo-2-1B-I	Values only	18.8	19.8	18.3	-2.7 [-5.0, -0.5]	-1.7 [-4.0, +0.7]

Table 19: Numeric-equation and name-repetition controls. Queried/intermediate columns rename that variable. Added errors subtract the ordinary twin’s suggested-digit rate (percentage points). Parse failures remain in denominators; coverage is 99.2–100% (rounded).

Model	Query name	Answer format	Pairs	Ordinary correct	Misleading correct	Added errors [95%]
Llama-3.2-3B	Omitted	Digit	600	99.7	99.8	+0.0 [+0.0, +0.0]
Llama-3.2-3B	Omitted	Assignment	600	100.0	100.0	+0.0 [+0.0, +0.0]
Llama-3.2-3B	Repeated	Digit	600	100.0	100.0	+0.0 [+0.0, +0.0]
Llama-3.2-3B	Repeated	Assignment	600	100.0	100.0	+0.0 [+0.0, +0.0]
Llama-3.1-8B	Omitted	Digit	600	100.0	100.0	+0.0 [+0.0, +0.0]
Llama-3.1-8B	Omitted	Assignment	600	100.0	100.0	+0.0 [+0.0, +0.0]
Llama-3.1-8B	Repeated	Digit	600	100.0	100.0	+0.0 [+0.0, +0.0]
Llama-3.1-8B	Repeated	Assignment	600	100.0	100.0	+0.0 [+0.0, +0.0]
OLMo-2-1B-I	Omitted	Digit	368	78.0	69.0	+12.8 [+9.1, +16.6]
OLMo-2-1B-I	Omitted	Assignment	368	94.0	94.3	+0.0 [+0.0, +0.0]
OLMo-2-1B-I	Repeated	Digit	368	96.2	94.3	+0.5 [+0.0, +1.4]
OLMo-2-1B-I	Repeated	Assignment	368	96.2	96.5	+0.0 [+0.0, +0.0]

Table 20: Final answers after fixed, completely correct calculations, with the misleading name on the queried variable. Correct answers are percentages; added errors are percentage points. All continuations parse. Assignment outputs include the queried name before the generated digit.

N Scope checks

Operation and scale. Moving the misleading name to a unary middle step produces no reliable effect for models above the accuracy floor; the largest excess is +0.6 points. CodeLlama-34B’s length-to-sum effect is +0.0, while its maximum-to-sum effect is +1.8 points [+0.7, +3.1]. The latter is a reliable positive effect, but its interval extends beyond the two-point margin. Its small point estimate cannot be reported as equivalence. The larger-checkpoint result supports the registered maximum-to-sum direction and reliable-effect criterion, while its magnitude is much smaller than CodeLlama-7B’s.

Natural code. We select 89 CRUXEval functions with length, loop-counter or maximum assignments, rename the target by AST rewriting to *v* or *total*, and verify execution after renaming. Direct and prose accuracy contrasts all have intervals covering zero on the 84 functions whose

renamed target is used downstream. Their endpoints stay within 10 and 13 points, respectively (Table 29). These are bounded results for those functions and prompts; they do not test the trace that lists values without expressions used in the synthetic study. The natural-code functions come from Gu et al. (2024).

O Effect and equivalence checks

Each contrast subtracts the neutral twin’s rate for the same digit. We resample matched problems with demonstration-seed replicates together and use 95% cluster-bootstrap intervals. Reliability also requires a consistent sign across all three demonstration sets.

Equivalence applies the ± 2 -point margin to 90% intervals from the same cluster bootstrap (two one-sided 5% tests), using 2,000 draws. Across 192 tested contrasts, 150 intervals lie wholly inside the margin, 36 contrasts have reliable effects without meeting that bound, and 6

Level	Neutral program (body)	Traced steps
1	<code>v = len(xs); return v</code>	1
2	<code>v = len(xs); w = v * 2; return w</code>	2
3	<code>v = len(xs); w = max(xs); u = v + w; return u</code>	3
4	level 3 with an unused fourth assignment	3
5	<code>v = len(xs); w = v * 2; u = w + 3; return u</code>	3
6	level 5 with the misleading name on the middle step	3

Table 21: The levels, one example each. The aggregate in the first statement is any of `len`, `sum`, `max`, `min`; unary operations are `*2`, `//2`, `+c`, `-c`. The misleading name sits on the first statement at levels 1–5 and on the second at level 6.

Family	Names	Implied operation
<code>len</code>	<code>count</code> , <code>n_items</code> , <code>length</code>	<code>len(xs)</code>
<code>sum</code>	<code>total</code> , <code>sum_all</code> , <code>acc</code>	<code>sum(xs)</code>
<code>max</code>	<code>largest</code> , <code>max_val</code> , <code>peak</code>	<code>max(xs)</code>
<code>min</code>	<code>smallest</code> , <code>min_val</code> , <code>low</code>	<code>min(xs)</code>
<code>double</code>	<code>double</code> , <code>twice</code>	<code>2 * a</code>
<code>half</code>	<code>half</code>	<code>a // 2</code>
<code>succ</code>	<code>succ</code> , <code>next_val</code>	<code>a + 1</code>
<code>pred</code>	<code>pred</code> , <code>prev_val</code>	<code>a - 1</code>
<code>add</code>	<code>combined</code> , <code>both</code>	<code>a + b</code>
<code>sub</code>	<code>diff</code> , <code>gap</code>	<code>a - b</code>
<code>mul</code>	<code>product</code> , <code>scaled</code>	<code>a * b</code>

Table 22: Identifier families and their suggested operations. The gate (Appendix J) scores each name by the log-probability each model assigns to the candidate right-hand sides; 12 of 25 names are read as intended by at least 80% of models. The `min` and `max` names cannot produce name errors on the aggregate step (their value is an input element) and are used as congruent and alternative-name names.

meet neither criterion. A small estimate or an interval covering zero does not by itself establish equivalence.

P Generated-prefix audit and initial readouts

For 2565 of 2565 misleading-name generations, the token prefix through the marker before the first value exactly matches the cached trace. This verifies that the cached state at that marker also precedes the generated value. No target value token is included. These initial readouts select layers on the full neutral evaluation set; the main text instead uses independent cross-fitted selection (Appendix S). We split the generated writes into name-suggested values, correct values and other errors. Table 30 averages three probe seeds and pools demonstration sets 7, 11 and 13. Intervals resample matched programs with all their demonstration repeats (4,000 draws). These conditional readouts establish available information on observed errors; they do not establish its causal use.

Q Sensitivity to identifier meanings

None of the three main-cell names (`total`, `sum_all`, `acc`) meets the originally planned 80% panel gate. Its filtered estimate is therefore unavailable, rather than zero. Table 31 retains the full-data estimate and reports each name and each model’s own meaning gate. Gates use the original saved meaning checks. Intervals cluster matched programs across the three demonstration sets. This exploratory sensitivity does not replace the planned panel gate or remove the deviation from it.

R Length and numeric elaboration controls

We add neutral annotations to each demonstrated value, matching the expression demonstrations’ token counts, and separately add valid numeric elaborations of the form `name = value + 0 = value`. The test program and Trace: prefix are unchanged. Greedy decoding uses the same 285 misleading programs and neutral twins for each format. Table 32 pools three demonstration sets; changes relative to the original trace and expression trace are paired on the same programs. The

Model	Return acc.	Direct: errors	Trace: len→sum	Trace: max→sum
OLMo-2-7B	95.7	+0.4	+58.4* [+53.0, +63.5]	+1.9 [+1.1, +3.0]
Llama-3.2-3B	97.4	+0.8	+43.4* [+38.5, +48.1]	+0.0 [+0.0, +0.0]
Llama-3.1-8B	99.6	+0.5	+12.4* [+8.8, +16.0]	+0.0 [+0.0, +0.0]
CodeLlama-7B	91.7	+1.0	+3.7* [+2.2, +5.4]	+10.7* [+7.9, +13.7]
CodeGemma-7B	99.2	+1.5*	+0.1 [+0.0, +0.4]	+0.0 [+0.0, +0.0]
Gemma-3-4B	99.1	+0.8	+0.0 [+0.0, +0.0]	+0.0 [+0.0, +0.0]
OLMo-2-1B	37.8	+3.1*	+3.7* [+2.2, +5.5]	+4.2* [+2.6, +5.9]
CodeLlama-34B	98.9	–	+0.0 [+0.0, +0.0]	+1.8* [+0.7, +3.1]

Table 23: Full level-5 panel: ordinary-name final-answer accuracy (%) and excess name errors (points). Direct contrasts rename the returned variable; trace contrasts rename the first assignment. Trace columns include 95% intervals. Stars mark consistent signs and intervals excluding zero.

Model	len→sum	max→sum	min→sum	sum→len	max→len
OLMo-2-7B	+17.6* / +58.4*	+0.9 / +1.9	+0.0 / +0.2	+4.4* / +0.0	+0.0 / +0.0
Llama-3.2-3B	+23.6* / +43.4*	+1.1 / +0.0	+0.0 / +0.0	+0.0 / +0.0	+0.0 / +0.0
Llama-3.1-8B	+1.5 / +12.4*	+0.7 / +0.0	+0.0 / +0.0	+0.0 / +0.0	+0.0 / +0.0
CodeLlama-7B	+0.9 / +3.7*	+7.5 / +10.7*	+1.0 / +1.7	+1.5 / +0.0	+0.9 / +0.6
CodeGemma-7B	+0.0 / +0.1	+10.8* / +0.0	+0.0 / +0.0	+0.0 / +0.0	+0.0 / +0.0
Gemma-3-4B	+0.0 / +0.0	+0.2 / +0.0	+0.0 / +0.0	+0.0 / +0.0	+0.0 / +0.0
OLMo-2-1B	+1.2 / +3.7*	+2.8* / +4.2*	+0.3 / -0.2	+10.6* / +11.1*	+9.6* / +5.6*

Table 24: Excess name errors at the aggregate step, trace regime, level 3 / level 5, three sets of worked examples pooled; * as in Table 23.

length-matched annotation controls retain much larger excess than expression demonstrations in both models. Numeric elaboration reduces excess relative to the original trace but leaves a larger effect than source expressions. Token count alone does not reproduce the expression benefit, and adding valid arithmetic accounts for part of it; these two contrasts do not isolate a single mechanism.

S Independent probe layer selection

The main-text error readouts use five-fold cross-fitting. A fixed SHA256 hash assigns canonical program computations to folds, keeping renamed versions and demonstration repeats together. For each held-out fold, we select the layer with the highest neutral accuracy on the other four folds, averaging three probe seeds and breaking ties toward the lowest layer. Neither the held-out neutral labels nor its misleading writes enter that choice. Probe training remains disjoint from every evaluation computation. Each program is scored only with its held-out fold’s layer. The generated-prefix audit from Appendix P still applies. Table 33 reports the conditional write strata and paired changes from the earlier full-neutral selection. Intervals use 4,000 canonical-program bootstrap draws, keeping demonstration repeats together. They condition on the fixed fold selectors

and do not refit layer selection inside each bootstrap. Name controls and the full layer curves remain descriptive measurements in Appendix M.

T Independent replication with screened names

We fix 16 fresh sum-family candidates before scoring them with the original candidate-continuation meaning check (Appendix J). The panel is the same seven models used for that gate. A name qualifies when sum is the highest-scoring continuation in at least 80% of the panel; ties follow the original scoring order. Two additional function-name contexts and neutral names are diagnostic and do not affect selection. Of the candidates, 14 qualify. We freeze the first 3 qualifying names in the fixed order: `sum_of_values`, `sum_of_items`, `sum_of_elements`. No trace outcomes enter selection. The replication uses 200 new length-computed programs. Their canonical computations are unique and absent from the original level-5 training and test corpus and all three demonstration sets. True and name-suggested values are checked by execution and retain the original value-exclusion rules. Every selected name is crossed with every program; its neutral twin differs only in the target identifier. The three affected models and the prespecified CodeGemma-7B control use the same programs and demonstration sets

Example format	Shown value	OLMo-2-7B-I	Llama-3.2-3B-I	Llama-3.1-8B-I
Values only	<code>v = 2</code>	+58.4	+43.4	+12.4
Matched descriptive text	<code>v = 2 [notes]</code>	+56.6	+49.6	–
Numeric elaboration	<code>v = 2 + 0 = 2</code>	+41.3	+14.9	–
Interactive Python	<code>>>> v 2</code>	+9.5	+12.6	+0.2
Code comments	<code>v = len(xs) # 2</code>	+0.0	+6.8	+0.1
Expression and value	<code>v = len(xs) = 2</code>	+3.4	+0.4	+0.0

Table 25: Code format comparison: added sum writes (points). Dashes: untested.

Format shown in examples	Example	OLMo-2-7B	Llama-3.2-3B	Llama-3.1-8B
Values only	<code>v = 2</code> <code>>>> v</code>	+58.4* [+53.0, +63.5]	+43.4* [+38.5, +48.1]	+12.4* [+8.8, +16.0]
Interactive Python transcript	<code>2</code>	+9.5* [+6.9, +12.0]	+12.6* [+9.4, +15.9]	+0.2 [+0.0, +0.6]
Code with value comments	<code>v = len(xs) # 2</code>	+0.0 [+0.0, +0.0]	+6.8* [+4.4, +9.2]	+0.1 [+0.0, +0.4]
Expression and value	<code>v = len(xs) = 2</code>	+3.4* [+2.0, +5.1]	+0.4 [+0.0, +0.9]	+0.0 [+0.0, +0.0]

Table 26: Baseline formats with 95% bootstrap intervals over matched problems.

7, 11 and 13 under trace and expression demonstrations. Table 35 pools names and demonstration sets, clustering all repeats of each computation in 4,000 bootstrap draws. Table 36 reports each identifier. The format contrast replicates on independently selected names, but names and programs differ from the original experiment, so differences in effect magnitude across the two samples are descriptive. This exploratory replication does not retroactively repair the original selection-rule deviation. Probes and patches concern the original identifiers, not these new names.

U Prompt-end versus pre-write readouts

We fit separate probes at the final token of the complete prompt ending in `Trace:`, before generating any trace token. Tokenization of this prefix is checked against the original continuation. Training computations, demonstration sets, SGD recipe and seeds match the pre-write probes. Each position selects its layer independently using the same five canonical-computation folds and other folds’ neutral labels. Outcome strata and generated-prefix eligibility are frozen from the previous audit. Intervals use 4,000 paired canonical-computation bootstrap draws and condition on the fixed selectors.

V Reproducibility and model terms

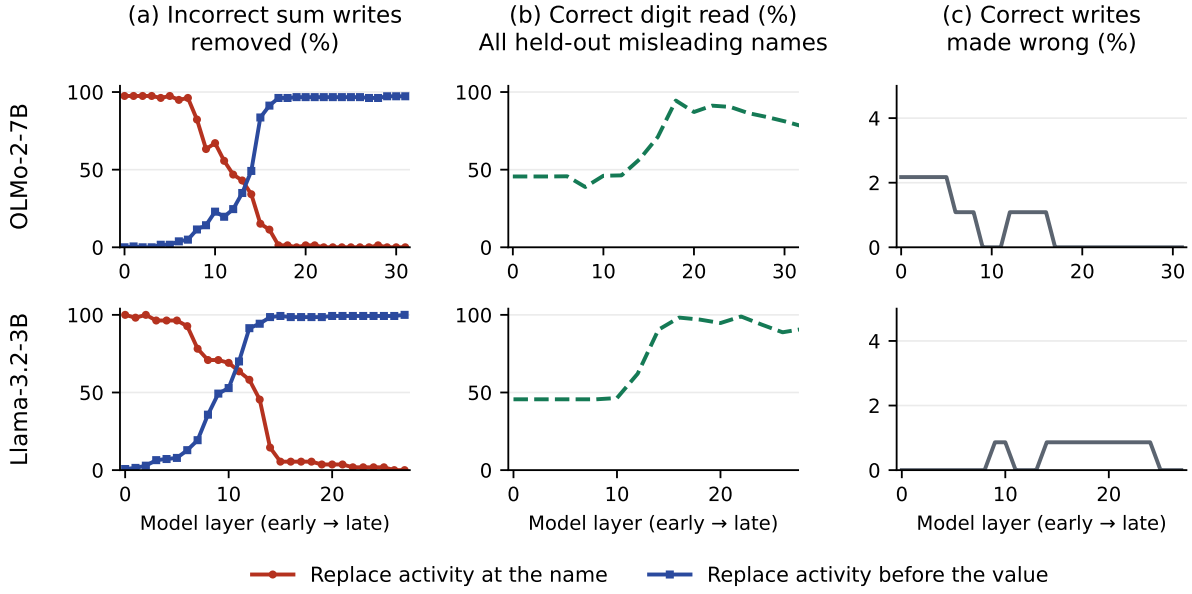
The main panel contains seven instruction-tuned checkpoints, with CodeLlama-34B as a scale check. We use CodeLlama (Rozière et al., 2023), CodeGemma (CodeGemma Team, 2024), Llama 3 (Grattafiori et al., 2024), Gemma 3

(Gemma Team, 2025) and OLMo 2 (Team OLMo, 2024; Walsh et al., 2025). The model identifiers are `codellama/CodeLlama-7b-Instruct-hf` and its 34b variant, `google/codegemma-7b-it`, `meta-llama/Llama-3.2-3B-Instruct`, `meta-llama/Llama-3.1-8B-Instruct`, `google/gemma-3-4b-it`, `allenai/OLMo-2-0425-1B-Instruct` and `allenai/OLMo-2-1124-7B-Instruct`. These use the Llama 2/CodeLlama, Llama 3.1 or 3.2 Community Licenses, Gemma Terms of Use, and Apache 2.0, respectively. The study uses their released weights for research inference and does not redistribute weights. CRUXEval uses the MIT license.

Synthetic data consist of digits, generated identifiers and short Python functions, without personal records or human annotations. Each tested level has 1,000 matched problems per target. Main behavior uses three sets of three demonstrations with seeds 7, 11 and 13; programs and demonstration identities remain fixed across compared formats. Greedy decoding uses the full vocabulary. Parsers retain failed outputs in denominators; trace-error measurements use the first assignment, while final-answer accuracy scores the reported return value. Probe training and testing are disjoint in arithmetic computation, ignoring identifier spelling. The independent-name replication uses separate screened names and 200 new computations, as described in Appendix T.

Probes use full-batch SGD with learning rate 10^{-3} for 10,000 epochs and seeds 0, 1 and 2. Five-fold layer selection is disjoint in computation; bootstrap intervals condition on those selectors.

Replacing internal activity can change wrong sum writes



Replacement uses the ordinary-name twin; removing a sum write can produce another wrong value.

Figure 9: **(a)** Name-error removal. **(b)** Correct-digit readout. **(c)** Damage to correct writes. Patches use ordinary-name twins.

These are linear-predictor fits, not language-model training. Combined recorded allocations across the arithmetic and code experiments total 222.5 GPU-hours on NVIDIA A30, H100 and H200 GPUs, an upper bound including idle and shared workers. Software is PyTorch 2.11.0 (CUDA 12.9) (Paszke et al., 2019), Transformers 5.14.1 (Wolf et al., 2020), NumPy 2.3.5 and scikit-learn 1.9.1 (Pedregosa et al., 2011).

Model	Set	Layer	Neutral	Control	Misleading	Control	Majority
OLMo-2-7B	7	18	99.5	100.0	94.5	33.0	45.6
OLMo-2-7B	11	18	97.3	100.0	95.4	0.0	45.6
OLMo-2-7B	13	18	99.8	100.0	97.4	23.4	45.6
Llama-3.2-3B	7	24	98.8	99.7	93.7	20.0	45.6
Llama-3.2-3B	11	20	100.0	99.8	95.4	5.0	45.6
Llama-3.2-3B	13	28	99.4	100.0	99.9	1.9	45.6
Llama-3.1-8B	7	16	100.0	87.8	99.9	33.0	45.6
Llama-3.1-8B	11	20	100.0	81.6	100.0	0.1	45.6
Llama-3.1-8B	13	20	100.0	87.2	100.0	3.0	45.6

Table 27: Disjoint code probes, three demonstration sets. Neutral and misleading are true-digit accuracy; each control column gives name-identity accuracy for that condition. Majority predicts the most frequent digit in training on the misleading test programs. Rates are percentages, averaged over three probe seeds. Layers are selected by neutral accuracy.

Model	Name tokens			Decision token			
	L0	damage	<10% by	50% at	90% at	all layers	damage
OLMo-2-7B	97%	2%	L17	L15	L16	97%	2%
Llama-3.2-3B	100%	0%	L15	L10	L12	100%	0%
Llama-3.1-8B	–	–	–	L8	L9	100%	0%

Table 28: Layers behind Figure 9. Name tokens: name errors removed by replacing every token of the name at layer 0, damage to correct instances, and the first layer after which removal falls below 10%. Decision token: first layer at which a single-layer replacement removes 50% and 90% of name errors, removal with every layer replaced, and its damage. Llama-3.1-8B’s name errors fall on a two-token name without a token-aligned twin.

Model	Direct acc.		Direct Δ	Prose acc.		Prose Δ
OLMo-2-7B	10	+1.2	[+0.0, +3.6]	22	-1.2	[-9.5, +8.3]
Llama-3.2-3B	19	+1.2	[-2.4, +4.8]	25	+3.6	[-3.6, +10.7]
Llama-3.1-8B	22	-3.6	[-9.5, +2.4]	40	+0.0	[-9.5, +9.5]
CodeLlama-7B	18	+2.4	[+0.0, +6.0]	13	+2.4	[-4.8, +9.5]
CodeGemma-7B	18	+0.0	[-4.8, +4.8]	22	+2.4	[-8.3, +13.1]
Gemma-3-4B	15	-2.4	[-6.0, +0.0]	45	+4.8	[-1.2, +11.9]
OLMo-2-1B	16	-3.6	[-8.3, +0.0]	3	+6.0	[+0.0, +11.9]

Table 29: 89 CRUXEval functions that assign a variable from `len(. . .)` (48), a loop counter (38) or `max(. . .)` (3), of which 84 use it downstream; each rendered with the original name, the neutral name `v`, and the misleading name `total`. Accuracy on all neutral renderings, and the paired misleading-minus-neutral difference in points on the downstream-used subset, with a bootstrap interval over functions.

Model	Generated write	Rows	Programs	True-digit accuracy (%)
OLMo-2-7B	name error	503	193	96.4 [94.8, 97.8]
OLMo-2-7B	correct write	320	135	96.7 [94.4, 98.5]
OLMo-2-7B	other error	32	24	77.1 [61.7, 90.5]
Llama-3.2-3B	name error	372	169	95.7 [93.8, 97.4]
Llama-3.2-3B	correct write	440	204	98.3 [97.0, 99.3]
Llama-3.2-3B	other error	43	29	82.2 [71.2, 91.7]
Llama-3.1-8B	name error	106	45	99.7 [99.0, 100.0]
Llama-3.1-8B	correct write	749	261	100.0 [100.0, 100.0]
Llama-3.1-8B	other error	0	0	–

Table 30: Disjoint probes at the verified generated prefix. A dash denotes an empty stratum. The denominator is observed writes within each stratum, not independent demonstration repeats.

Model	All	Model gate	total	sum_all	acc
Gemma-3-4B	0.0 [0.0, 0.0]	–	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
CodeGemma-7B	0.1 [0.0, 0.4]	0.1 [0.0, 0.4]	0.0 [0.0, 0.0]	0.3 [0.0, 1.0]	0.0 [0.0, 0.0]
Llama-3.1-8B	12.4 [8.9, 16.0]	12.4 [8.9, 16.0]	0.0 [0.0, 0.0]	34.0 [26.0, 42.3]	0.0 [0.0, 0.0]
OLMo-2-1B	3.7 [2.2, 5.5]	5.0 [1.8, 8.9]	5.0 [1.8, 8.9]	5.8 [2.6, 9.3]	0.0 [0.0, 0.0]
OLMo-2-7B	58.4 [53.1, 63.4]	59.3 [52.7, 65.8]	56.4 [47.9, 64.9]	96.2 [93.9, 98.1]	15.3 [9.6, 21.8]
CodeLlama-7B	3.7 [2.1, 5.5]	3.7 [2.1, 5.5]	1.8 [0.4, 3.2]	8.7 [4.5, 13.1]	0.0 [0.0, 0.0]
Llama-3.2-3B	43.4 [38.5, 48.0]	43.4 [38.5, 48.0]	30.1 [23.8, 36.9]	76.3 [70.2, 82.4]	18.4 [11.9, 25.7]

Table 31: Excess name errors in percentage points [95% interval], level 5 length-to-sum. Empty subsets are shown as dashes. The saved JSON also reports each demonstration set and the all-three-sets sign rule.

Model	Format	Excess	Change vs trace	Change vs expression
OLMo-2-7B	neutral annotation	56.6 [51.3, 61.6]	-1.8 [-3.4, -0.1]	53.2 [48.1, 58.2]
OLMo-2-7B	numeric elaboration	41.3 [36.4, 46.0]	-17.1 [-21.1, -12.9]	37.9 [33.2, 42.5]
Llama-3.2-3B	neutral annotation	49.6 [44.7, 54.3]	6.2 [4.1, 8.4]	49.2 [44.3, 53.9]
Llama-3.2-3B	numeric elaboration	14.9 [12.0, 17.8]	-28.5 [-32.7, -24.3]	14.5 [11.7, 17.4]

Table 32: Excess name errors and paired changes in percentage points [95% interval]. Bootstrap clusters keep all demonstration repeats for each matched program together. These controls assess length and numeric elaboration; they do not by themselves establish how expressions change generation.

Model	Write	Rows	Computations	Accuracy (%)	Change (points)
OLMo-2-7B	name error	503	193	96.4 [94.9, 97.9]	0.0 [0.0, 0.0]
OLMo-2-7B	correct write	320	135	96.7 [94.4, 98.4]	0.0 [0.0, 0.0]
OLMo-2-7B	other error	32	24	77.1 [61.6, 90.6]	0.0 [0.0, 0.0]
Llama-3.2-3B	name error	372	169	95.5 [93.6, 97.3]	-0.2 [-0.9, 0.4]
Llama-3.2-3B	correct write	440	204	98.0 [96.6, 99.1]	-0.3 [-0.8, 0.0]
Llama-3.2-3B	other error	43	29	82.2 [71.1, 91.7]	0.0 [0.0, 0.0]
Llama-3.1-8B	name error	106	45	99.7 [99.0, 100.0]	0.0 [0.0, 0.0]
Llama-3.1-8B	correct write	749	261	100.0 [99.9, 100.0]	-0.0 [-0.1, 0.0]
Llama-3.1-8B	other error	0	0	–	–

Table 33: True-digit readouts with independent layer selection. Changes compare the same observed writes with the earlier full-neutral layer choice. A dash denotes an empty stratum. These readouts establish decodability, not causal use.

Candidate	Sum agreement	Qualifies	Selected
sum_of_values	6/7	yes	yes
sum_of_items	7/7	yes	yes
sum_of_elements	7/7	yes	yes
sum_of_xs	7/7	yes	no
sum_of_list	6/7	yes	no
list_sum	6/7	yes	no
input_sum	6/7	yes	no
items_sum	6/7	yes	no
elements_sum	5/7	no	no
sum_values	5/7	no	no
sum_result	6/7	yes	no
sum_total	6/7	yes	no
total_sum	6/7	yes	no
xs_sum	7/7	yes	no
sum_xs	7/7	yes	no
summation	6/7	yes	no

Table 34: The entire fixed candidate list, in selection order. Qualification uses the original canonical meaning prompt. Selection retains the first three qualifying candidates without consulting behavior.

Model	Trace excess	Expression excess	Expression minus trace
OLMo-2-7B	91.1 [88.4, 93.6]	0.8 [0.3, 1.3]	-90.3 [-92.9, -87.6]
Llama-3.2-3B	61.5 [56.1, 66.9]	0.0 [0.0, 0.0]	-61.5 [-66.9, -56.1]
Llama-3.1-8B	16.7 [12.8, 20.8]	0.0 [0.0, 0.0]	-16.7 [-20.8, -12.8]
CodeGemma-7B	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]

Table 35: Independent screened-name replication: excess name errors and paired format changes in percentage points [95% interval]. All selected identifiers and three demonstration sets are pooled over 200 fresh computations. A zero bootstrap interval records no observed paired name errors; it is not a population-level absence or equivalence claim.

Model	Identifier	Trace excess	Expression excess
OLMo-2-7B	sum_of_values	91.7 [88.8, 94.3]	0.2 [0.0, 0.5]
OLMo-2-7B	sum_of_items	89.2 [86.0, 92.0]	2.0 [0.8, 3.3]
OLMo-2-7B	sum_of_elements	92.5 [89.8, 94.8]	0.2 [0.0, 0.5]
Llama-3.2-3B	sum_of_values	61.7 [56.0, 67.3]	0.0 [0.0, 0.0]
Llama-3.2-3B	sum_of_items	55.5 [50.0, 61.3]	0.0 [0.0, 0.0]
Llama-3.2-3B	sum_of_elements	67.3 [61.7, 73.0]	0.0 [0.0, 0.0]
Llama-3.1-8B	sum_of_values	12.8 [9.3, 16.7]	0.0 [0.0, 0.0]
Llama-3.1-8B	sum_of_items	22.8 [18.0, 27.7]	0.0 [0.0, 0.0]
Llama-3.1-8B	sum_of_elements	14.5 [10.5, 18.7]	0.0 [0.0, 0.0]
CodeGemma-7B	sum_of_values	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
CodeGemma-7B	sum_of_items	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
CodeGemma-7B	sum_of_elements	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]

Table 36: Per-identifier replication, pooling the same three demonstration sets. Effects are percentage points [95% canonical-program bootstrap interval]. All affected-model trace means are positive in every demonstration set; per-set estimates are retained in the saved summary.

Model	Write	Rows / computations	Prompt	Before write	Change	Name ID
OLMo-2-7B	sum write	503 / 193	38.4 [32.9, 43.9]	96.4 [94.9, 97.9]	+58.1 [+52.6, +63.4]	12.7 [9.8, 15.9]
OLMo-2-7B	correct write	320 / 135	35.2 [27.7, 43.1]	96.7 [94.4, 98.4]	+61.5 [+53.3, +69.3]	20.8 [15.5, 26.3]
OLMo-2-7B	other error	32 / 24	76.0 [58.1, 90.2]	77.1 [61.6, 90.6]	+1.0 [-19.6, +22.6]	26.0 [12.6, 41.1]
Llama-3.2-3B	sum write	372 / 169	26.4 [22.4, 30.8]	95.5 [93.6, 97.3]	+69.1 [+64.5, +73.4]	0.7 [0.1, 1.6]
Llama-3.2-3B	correct write	440 / 204	37.9 [32.1, 43.7]	98.0 [96.6, 99.1]	+60.1 [+54.4, +65.8]	3.6 [1.8, 5.7]
Llama-3.2-3B	other error	43 / 29	20.9 [11.7, 31.7]	82.2 [71.1, 91.7]	+61.2 [+46.8, +73.8]	2.3 [0.0, 6.1]
Llama-3.1-8B	sum write	106 / 45	56.0 [47.4, 64.2]	99.7 [99.0, 100.0]	+43.7 [+35.4, +52.4]	0.0 [0.0, 0.0]
Llama-3.1-8B	correct write	749 / 261	49.9 [45.7, 54.1]	100.0 [99.9, 100.0]	+50.1 [+45.8, +54.3]	0.0 [0.0, 0.0]
Llama-3.1-8B	other error	0 / 0	—	—	—	—

Table 37: Paired true-digit accuracies and prompt-position name-identity controls (%) [95% interval]. Change is pre-write minus prompt accuracy in points.

Model	Extra sum writes (points)	Correct digit (%) Before trace	Correct digit (%) Before wrong value	Wrong writes (unique programs)
OLMo-2-7B	+58.4	38.4 [32.9, 43.9]	96.4 [94.9, 97.9]	503 (193)
Llama-3.2-3B	+43.4	26.4 [22.4, 30.8]	95.5 [93.6, 97.3]	372 (169)
Llama-3.1-8B	+12.4	56.0 [47.4, 64.2]	99.7 [99.0, 100.0]	106 (45)

Table 38: Code behavior and error-conditioned readouts [95% interval]. Counts distinguish writes from computations.